

Review

Predictive Modeling for Electric Vehicle Battery State of Health: A Comprehensive Literature Review

Jianqiang Gong¹, Bin Xu^{2,*}, Fanghua Chen¹ and Gang Zhou¹

¹ Key Laboratory of Operation Safety Technology on Transport Vehicles, Research Institute of Highway Ministry of Transport, Beijing 100088, China; fh.chen@rioh.cn (F.C.); g.zhou@rioh.cn (G.Z.)

² College of Transportation Engineering, Chang'an University, Xi'an 710064, China

* Correspondence: xb216@chd.edu.cn

Abstract: The rising adoption of electric vehicles (EVs) utilizing lithium-ion batteries necessitates a robust understanding of state-of-health (SOH) estimation. The existing literature highlights various SOH estimation models, but a comprehensive comparative analysis is lacking. This paper addresses this gap by conducting an exhaustive review of diverse SOH estimation approaches for EV battery applications, including the direct measurement method, physical-based and data-driven approaches. Results highlight that data-driven methods, particularly those utilizing machine learning techniques, offer superior accuracy and adaptability but often require extensive datasets. In contrast, physical-based approaches provide interpretable insights but are computationally intensive, and direct measurement methods, though simple, lack generalizability. In addition, this paper also systematically reviews the indicators of battery SOH, influential factors affecting battery SOH, and various datasets used for SOH modeling. Future research should focus on integrating multiple modeling methodologies to leverage their combined strengths, enhancing the collection of comprehensive battery lifecycle datasets to support robust model development, and extending the scope of SOH estimation beyond individual cells to encompass entire battery packs.

Keywords: electric vehicles; state of health; estimation; comparative analysis



Academic Editor: José Gabriel Oliveira Pinto

Received: 30 November 2024

Revised: 30 December 2024

Accepted: 10 January 2025

Published: 14 January 2025

Citation: Gong, J.; Xu, B.; Chen, F.; Zhou, G. Predictive Modeling for Electric Vehicle Battery State of Health: A Comprehensive Literature Review. *Energies* **2025**, *18*, 337. <https://doi.org/10.3390/en18020337>

Copyright: © 2025 by the authors. Licensee MDPI, Basel, Switzerland. This article is an open access article distributed under the terms and conditions of the Creative Commons Attribution (CC BY) license (<https://creativecommons.org/licenses/by/4.0/>).

1. Introduction

The transportation sector is a significant contributor to greenhouse gas emissions, accounting for roughly 14% of global greenhouse gas emissions in 2022 [1]. The transition to electric mobility has great potential to address the climate change challenges. Governments worldwide are developing ambitious targets and policies to incentivize electric vehicle (EV) adoption, signaling a paradigm shift away from internal combustion engine vehicles. The global EV market share reached 14% in 2022. Over 26 million electric cars were on the road in 2022, up 60% compared to 2021 and more than a fivefold growth since 2018 [2].

Lithium-ion batteries, serving as the primary energy storage units, constitute essential components in EVs thanks to their high energy and power density and long lifecycle [3,4]. The EV battery requires a battery management system (BMS) to guarantee safe and reliable EV operation. The BMS can provide real-time feedback of battery status information, where SOH is one of the most important measures to be monitored [5]. Accurate estimation for the battery SOH informs the timing for replacing any malfunctioning batteries, keeping the EV battery in optimal operating condition, and decreasing the costs associated with using EVs.

Although there have been several literature review papers on EV battery SOH estimation, they have the following limitations. Berecibar et al. (2016) reviewed battery SOH monitoring techniques [6]. However, this literature review was not comprehensive and summarized papers that focused on early generations of EVs. Xiong et al. (2018) categorized the battery SOH estimation approach into experimental and physical-based methods, but the advantages and disadvantages of each estimation method were not thoroughly discussed, which is critical for understanding their practical applicability and limitations [7]. Ungurean et al. (2017) and Lipu et al. (2018) reviewed the estimation approaches for battery SOH [8,9]. However, both reviews failed to explore the influential factors affecting battery SOH, such as temperature, state of charge, and cycling conditions, which are pivotal for accurate SOH modeling. Li et al. (2019) introduced the data-driven methods for battery SOH estimation, including machine learning and differential analysis. Although their review highlighted the promise of data-driven techniques, it did not consider other widely used approaches, such as electrochemical models and hybrid frameworks, thereby providing an incomplete perspective of the field [10]. A summary of existing literature review papers on SOH modeling is shown in Table 1.

These gaps in the existing literature indicate a lack of a unified framework that comprehensively evaluates the various SOH estimation approaches while considering influential factors, key indicators, and data requirements. Motivated by the research gaps, this paper aims to provide a comprehensive review that addresses these deficiencies and synthesizes existing knowledge into actionable insights for future research and practical applications. To this end, the paper contributes to the existing literature in three ways:

- (1) systematically reviews EV battery SOH estimation methods, covering direct measurement, physics-based, data-driven, and hybrid approaches, with a focus on their strengths, limitations, and appropriate application scenarios;
- (2) identifies and categorizes key SOH indicators and examines influential factors of battery SOH, such as environmental conditions, usage patterns, and operational variables, that have been underexplored in previous reviews;
- (3) reviews available datasets and highlights their importance in the development and validation of SOH estimation models.

This paper is organized as follows. Section 2 introduces the literature screening method. Li-ion battery working principles are summarized in Section 3. The indicators of battery SOH are reviewed in Section 4. Section 5 focuses on the influential factors of battery SOH. Various datasets used for battery SOH modeling are detailed in Section 6. Different battery SOH modeling approaches are provided in Section 7. Lastly, the key limitations and future research avenues are discussed in Section 8. Section 9 concludes this paper with final remarks.

Table 1. Summary of existing literature review papers on SOH modeling [6,7,9,10].

Study	Time Period of Covered Literature	Contribution	Limitations	Main Findings
Berecibar et al. (2016) [6]	1983–2015	A comprehensive review was conducted on various methods for estimating state of health (SOH) in lithium-ion batteries, analyzing their advantages and disadvantages. A new approach combining degradation mechanism detection and adaptive models or big data methods was proposed to improve the accuracy and practicality of SOH estimation.	Accuracy, computational complexity, model complexity and parameter sensitivity, robustness and reliability will be affected	Pointing out experimental methods such as direct measurement and measurement-based models. Adaptive models such as Kalman filters, observers, fuzzy logic, and artificial neural networks were mentioned. Although some methods have high accuracy, they often bring high computational costs, which makes their implementation challenging. A method using differential voltage curves to detect degradation mechanisms was proposed to improve SOH estimation by providing insights into the battery aging process. Combining adaptive models or big data with degradation mechanism detection may develop more accurate and practical SOH estimation methods.

Table 1. Cont.

Study	Time Period of Covered Literature	Contribution	Limitations	Main Findings
Xiong et al. (2018) [7]	1993–2018	A comprehensive critical review of battery state of health (SOH) monitoring methods was conducted, with detailed classification and explanation of various SOH estimation methods. The advantages and disadvantages of these methods were analyzed in depth, and the shortcomings and improvement directions of existing research were pointed out. In addition, the future development trends of SOH management were discussed, providing valuable reference and support for researchers and practitioners, and helping to promote the intelligent development of battery management systems.	Difficult to achieve real-time monitoring, high equipment requirements, and some destructive testing methods (such as Raman spectroscopy, X-ray diffraction, etc.) can permanently damage the battery. At present, most SOH monitoring methods mainly rely on external characteristics such as voltage, current, and temperature of the battery, and lack direct monitoring methods for the internal state of the battery. Although methods such as ultrasonic detection have potential, further research and improvement are still needed. Meanwhile, the calculations are quite complex.	The proposal of experimental methods and model-based methods, combined with multiple model fusion methods and the application of big data and machine learning technologies, is considered the development trend of SOH monitoring in the future, which is expected to improve the intelligence level of battery management systems and the safety of electric vehicles.
Lipu et al. (2018) [9]	2000–2018	A comprehensive review was conducted on the estimation methods of state of health (SOH) and remaining useful life (RUL) of lithium-ion batteries in electric vehicles. Existing SOH and RUL estimation methods were classified in detail, and various methods were compared and analyzed. Challenges and issues such as data acquisition difficulty, model complexity, and computational cost were explored in depth, and a series of recommendations and suggestions were proposed, including improving data acquisition and processing methods, developing more efficient algorithms and models, and strengthening interdisciplinary cooperation.	Data may be scarce and of low quality, with weak environmental adaptability and limited coverage	The performance of lithium-ion batteries is influenced by various internal and external factors, such as material properties, thermal runaway, capacity decay, overcharging and overdischarging, lifecycle, hysteresis effect, aging, charge discharge rate, and uncertainty of battery models. External factors such as charging methods, safety and protection, cost, environmental impact, and recycling also pose challenges to the prediction of SOH and RUL.

Table 1. Cont.

Study	Time Period of Covered Literature	Contribution	Limitations	Main Findings
Li et al. (2019) [10]	2000–2019	A comprehensive review of data-driven methods for health estimation and lifespan prediction of lithium-ion batteries was conducted, and the advantages and disadvantages of different methods were classified and discussed. The feasibility and cost-effectiveness of data-driven methods in practical applications were emphasized, and the challenges and future development directions faced by data-driven methods were discussed. Practical technical comparisons and selection guidelines were provided, and the broad prospects of data-driven technology in the field of battery research were discussed.	Dependent on data, poor generalization ability, complex calculations, sensitive operations, model difficulties, limited early prediction ability	The aging of lithium batteries is mainly caused by three main mechanisms: lithium inventory loss (LLI), active material loss (LAM), and increased internal resistance of the battery. The advantages and limitations of various methods such as differential analysis, machine learning, and analytical models in battery health diagnosis and prediction were pointed out. Although the differential analysis method is easy to implement and has a small computational cost, it is greatly affected by testing conditions such as temperature and current rate. Although machine learning methods can handle complex non-linear systems and provide high estimation accuracy, they are computationally intensive and highly dependent on the quality and quantity of training data. Although analytical modeling methods can provide relatively stable prediction results, their generalization ability is weak and difficult to adapt to different operating conditions. Future research should focus on combining physical mechanisms with data-driven methods to improve the accuracy and applicability of predictions.

2. Literature Screening Methods

To ensure the comprehensiveness of this literature review, a systematic approach was adopted to identify relevant studies. Key academic databases, including Scopus, Web of Science, and Google Scholar, were queried. First, a combination of keywords and Boolean operators were used to construct the search strings, including terms such as “electric vehicle”, “Lithium-ion battery”, “State of health”, “SOH estimation”, and “predictive modeling”. Second, the following inclusion criteria were applied to select relevant literature: (1) papers published in peer reviewed journals within the last decade (from 2014 to 2024); (2) papers written in English; and (3) research focused on predictive modeling for electric vehicle battery health. In contrast, the exclusion criteria eliminated the following studies: (1) studies solely addressing non-electric vehicle batteries (e.g., stationary energy storage or portable devices); (2) studies lacking predictive modeling elements, such as studies solely discussing battery chemistry or experimental testing without modeling applications. Third, duplicate records identified across multiple databases were removed during the screening process. This was achieved using reference management software (i.e., Zotero). A manual cross-check was also performed to ensure accuracy in the deduplication process. Figure 1 shows the overall literature search procedure.

Literature Screening Procedure for EV Battery SOH Review

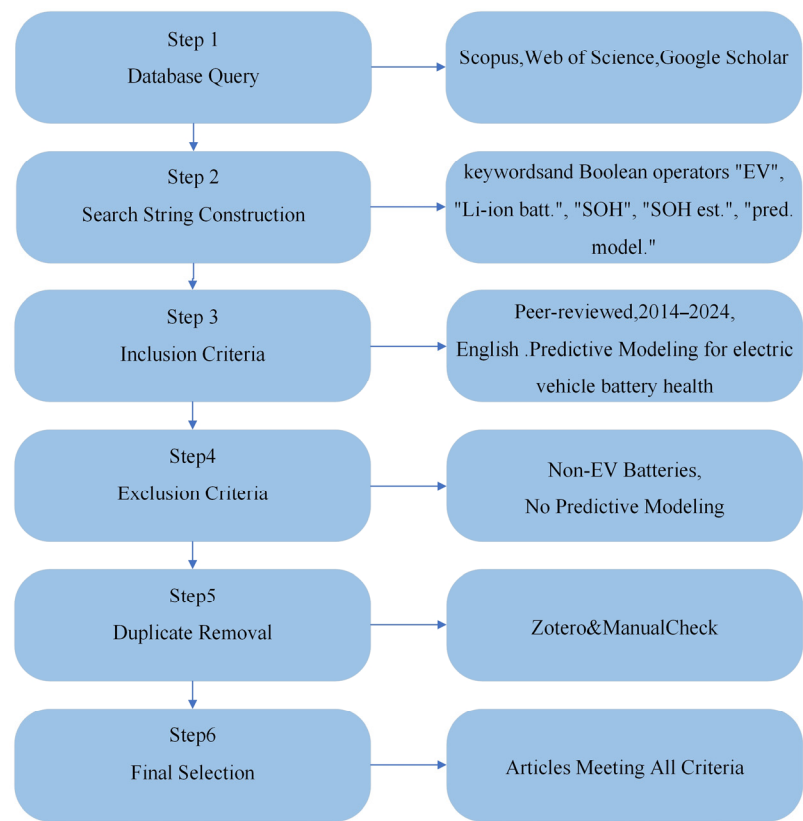


Figure 1. Literature screening procedure for EV battery SOH review.

3. Li-Ion Battery Working Principles

The structure of the Li-ion battery and the type of materials which are used in its components play a decisive role in its capacity, cycle, and safety performance. At present, the Li-ion battery is one of the most commonly used battery technologies. It has the characteristics of high energy density, long life, low self-discharge rate and suitability for electric vehicles or hybrid vehicles. However, lithium-ion batteries also have some

problems and challenges. Firstly, the charging time is relatively long, which limits their application in the field of fast charging. Secondly, due to the presence of organic solvents, it may catch fire or explode under high-temperature or short-circuit conditions. In addition, the current mainstream lithium-ion battery recycling technology is not perfect, and the disposal of waste batteries has become a difficult problem.

Currently, lithium-ion batteries are mainly divided into ternary lithium batteries and lithium iron phosphate batteries. Ternary lithium batteries have the advantage of high energy density which can provide greater driving range for new energy vehicles. Due to the chemical characteristics of metals such as nickel and cobalt, ternary lithium batteries have relatively high safety risks under extreme conditions such as high temperature and overcharge. Cobalt resources are scarce, resulting in high battery costs. The lithium iron phosphate battery has the following advantages: lifecycle is long; it is not easy to cause a thermal runaway phenomenon; the production process is simple; rich raw material resources are available; and the cost is relatively low. The disadvantage is that compared with ternary lithium battery, the energy density is relatively low. Based on the needs of high energy ratio, low cost and high safety, different transmission media, including lithium–sulfur batteries, sodium-ion batteries and lithium–air batteries, have been extensively studied and gradually loaded into vehicles for trial [11].

3.1. Composition of Lithium-Ion Batteries

Li-ion batteries are mainly composed of four key components: positive electrode materials, negative electrode materials, electrolytes and separator. The positive electrode materials typically consist of lithium compounds, such as lithium iron phosphate (LiFePO_4) or nickel manganese cobalt oxide (NMC). These materials are responsible for embedding or releasing lithium ions during the charge and discharge process. The negative electrode material is usually graphite, which has good electrical conductivity and lithium embedding performance. The negative electrode material is responsible for releasing or absorbing lithium ions during the charge and discharge process. Meanwhile, the positive electrode material is mutually embedded lithium. The electrolyte is the medium between the positive and negative electrodes, usually using organic solution or polymer gel. The electrolyte is responsible for the conduction of ions, so that lithium ions can move freely between the positive and negative electrodes, so as to achieve the charge and discharge function of the battery. The separator is located between the positive and negative electrodes to isolate the positive and negative electrodes and prevent short circuit. The separator is usually composed of polymer materials, which have good ion conductivity and isolation properties [12].

3.1.1. Cathode Materials

- Lithium Cobalt Oxide (LiCoO_2)

Lithium cobalt oxide was one of the earliest and most critical cathode materials for lithium-ion batteries, offering high capacity and good thermal stability. However, it is prone to thermal runaway under high temperatures and overcharging conditions, decomposing to release oxygen and posing significant safety risks [13]. Additionally, its capacity declines significantly under high-voltage cycling (>4.2 V) due to irreversible structural changes. Although surface coatings and electrolyte modifications have improved its stability, the high cost and limited cobalt resources restrict its large-scale application. In comparison, lithium nickel oxide (LiNiO_2) and manganese-based materials offer advantages in terms of cost and resource availability [14].

- Lithium Nickel Oxide (LiNiO_2)

LiNiO₂ exhibits a high theoretical capacity and is more cost-effective than LiCoO₂. However, its synthesis is challenging due to the tendency for nickel–lithium cation mixing, which degrades its cycling performance [15]. Doping strategies (e.g., Co, Al, Mn) have improved its structural stability and capacity retention. Compared to LiCoO₂, LiNiO₂ and its derivatives, such as NCA and NCM, deliver higher energy density at high voltage but face safety concerns associated with high nickel content. For instance, high-nickel NCM811 offers higher capacities but suffers from reduced cycle stability and thermal safety, limiting its broader application in electric vehicles [16].

- Manganese-Based Oxides

Manganese-based materials (e.g., LiMnO₂) are attractive for their low cost, environmental friendliness, and relatively high initial capacity. However, capacity decays rapidly during cycling, primarily due to the Jahn–Teller effect and structural transitions [17]. Solid solutions incorporating nickel and cobalt (e.g., LiNi_{0.5}Mn_{0.5}O₂) have been developed to improve cycling stability. Compared to lithium cobalt and nickel oxides, manganese-based materials are more suitable for low-power applications and are cost-effective candidates for large-scale energy storage systems [18].

- Phosphate Materials

Lithium iron phosphate (LiFePO₄), introduced in 1997, is a highly safe and long-lasting cathode material with excellent thermal stability and cycling performance. However, its low electronic conductivity limits its rate performance, which can be effectively improved through nanoscale engineering and carbon coating. Compared to LiCoO₂ and LiNiO₂, LiFePO₄ has lower energy density but is more affordable and thermally stable, making it ideal for long-life and high-safety applications, such as electric vehicles and energy storage systems [19].

- Iron Fluorides (FeF₃)

Iron fluorides have gained attention for their high theoretical capacity (>200 mAh/g) and thermal stability. However, their low electrochemical activity and poor cycling stability limit their practical applications. Nanocomposite fabrication (e.g., FeF₃/C composites) has significantly enhanced capacity retention. Compared to other cathode materials, the high-voltage characteristics of iron fluorides make them appealing for high-energy-density applications, but their limited cycling performance remains a significant challenge [20].

- Vanadium-Based Oxides (V₂O₅)

Vanadium-based oxides possess an open layered structure that enables high-lithium-ion-insertion/extraction capacity (~294 mAh/g). Despite their low cost and straightforward synthesis, their low lithium-ion diffusion rate and poor conductivity hinder cycling stability [21]. Nanostructure engineering, such as nanowires and hollow spheres, has effectively improved their electrochemical properties. Compared to LiFePO₄ and LiCoO₂, vanadium-based oxides offer higher capacity potential, but their rate performance and stability still require further optimization.

3.1.2. Anode Materials

- Carbon-Based Materials

Graphite is the most widely used anode material, offering high reversible capacity (372 mAh/g) and excellent cycling performance. However, its performance under high rates is limited, and lithium plating risks exist. Surface modifications and nanoscale engineering have improved its conductivity and interfacial stability. Compared to advanced materials like graphene and carbon nanotubes, graphite is more cost-effective and suitable for large-scale commercial applications [22].

- Carbon Nanotubes (CNTs)

CNTs are widely studied as high-performance anode materials due to their excellent electrical conductivity and mechanical strength. Composites incorporating metal oxides (e.g., $\text{Fe}_3\text{O}_4/\text{MWCNTs}$) or sulfides (e.g., MoS_2/CNTs) have significantly enhanced capacity and cycling life. However, challenges related to high production costs and scalability remain [23]. Compared to traditional graphite, CNTs exhibit higher energy density and better rate performance, but their application is constrained by manufacturing and economic challenges [24].

- Graphene

Graphene, with its high specific surface area and superior electrical conductivity, is a promising anode material. While its theoretical capacity reaches 1116 mAh/g, practical applications often suffer from agglomeration, leading to reduced capacity [25–28]. Functionalization and composite strategies have been employed to address these issues. Compared to CNTs, graphene offers better electrical properties and lithium storage capabilities but faces similar cost barriers for mass commercialization.

- Titanium-Based Oxides

Titanium-based oxides (e.g., TiO_2 and $\text{Li}_4\text{Ti}_5\text{O}_{12}$) are characterized by high safety and long lifecycle, making them ideal anode materials for electric vehicle applications. $\text{Li}_4\text{Ti}_5\text{O}_{12}$, in particular, avoids lithium dendrite formation due to its zero-strain characteristic and high operating voltage [29]. However, its low theoretical capacity limits its energy density. Compared to carbon-based materials, titanium-based oxides excel in safety and rate performance but require further research to enhance their capacity.

- Titanium-Based Derivatives

Different titanium oxide polymorphs (e.g., anatase, rutile, brookite) demonstrate varied performance in LIBs. Anatase TiO_2 , for example, offers higher capacity and rate performance but suffers from shorter lifecycle. Nanoscale engineering and surface modification have improved lithium-ion diffusion rates and electrical conductivity. Compared to other anode materials, titanium-based oxides provide outstanding safety and stability, though their practical capacity remains a focus of ongoing research [30].

3.1.3. Electrolytes

The electrolyte in the lithium battery is mainly composed of lithium salt and organic solvent, which are to ensure the ionic conduction between the positive and negative electrodes, so that the battery has the advantages of high voltage and high energy ratio. The basic components of the electrolyte include lithium salts, organic solvents and additives, which are formulated in specific proportions. The types of electrolytes used in lithium batteries mainly include liquid electrolytes, gel electrolytes and solid electrolytes [31,32].

3.1.4. Separator

The main material used in lithium battery separators is high-strength thin-film polyolefin porous membranes. This material usually has excellent physical and chemical properties, which can withstand organic solvents and has good electrolyte ion permeability and low ion resistance. The separator has the function of separating positive and negative terminals, allowing ions to pass through and improving the capacity, cycle performance and safety of the battery [33].

3.2. The Working Principle and Internal Chemical Reactions of Lithium-Ion Batteries

The charging and discharging mechanisms of lithium-ion battery (e.g., LiMnO_2) are essentially the cyclic movement of lithium ions between the positive and negative electrodes.

During charging, manganese ions are oxidized in the positive electrode and then broken down into lithium ions. The decomposed lithium ions pass through the intermediate electrolyte, and after passing through the diaphragm, they are combined with the electrons transmitted by the external circuit. Upon reaching the negative electrode, the lithium ions will be embedded in the pores of the carbon material. As the number of lithium ions embedded continues to increase, the charging capacity of lithium-ion batteries will also increase. During discharge, the lithium ions embedded in the negative carbon layer will escape back to the positive electrode and absorb electrons [34–37]. The internal chemical reactions during charging and discharging are represented as follows [38–41]:

Charging:

At the cathode: $\text{LiMO}_2 \rightarrow \text{Li}_{1-x}\text{MO}_2 + x\text{Li}^+ + xe^-$

At the anode: $x\text{Li}^+ + xe^- + \text{C} \rightarrow \text{Li}_x\text{C}$

Discharging:

At the cathode: $\text{Li}_{1-x}\text{MO}_2 + x\text{Li}^+ + xe^- \rightarrow \text{LiMO}_2$

At the anode: $\text{Li}_x\text{C} \rightarrow x\text{Li}^+ + xe^- + \text{C}$

4. Indicators of Battery State of Health

The SOH quantifies the degradation level of the battery over time and usage. It is often expressed as a percentage, where 100% indicates a fully healthy battery, and lower percentages reflect the degree of degradation. The battery SOH cannot be measured directly. Different indicators have been used to quantify the SOH, including the remaining capacity [42], internal resistance [43], and voltage [44,45], etc. Table 2 shows a comparative analysis for the indicators of battery SOH.

Table 2. A comparative analysis of the indicators for SOH.

SOH Indicator	Strength	Limitations	Applicable Scenarios
Internal resistance	- Directly reflects aging dynamics due to chemical and mechanical changes. - Simple to measure in controlled environments.	- Requires reference points (e.g., R_{eol}, R_0) which are difficult to standardize. - Highly sensitive to temperature and load fluctuations.	Laboratory experiments.
Capacity fading	- Intuitive and widely accepted in industry. - Represents a tangible loss of energy storage capacity.	- Dependent on full charge–discharge cycles, which are rare in real-world EV operations. - Affected by ambient temperature, current rate, and aging trajectory.	- EV dashboards showing the SOH of batteries. - Fleet management systems for monitoring battery usage.
CCDT, CCDC, MCCDR	- Can be monitored online during charging. - Provide indirect measures of SOH under real-world operational conditions.	- Require constant voltage charging stages, limiting applicability to specific scenarios.	- Scenarios where full cycling is impractical.

4.1. Internal Resistance

The internal resistance denotes the impedance encountered by the electric current within the battery, and its change over time is indicative of the battery's aging dynamics. As batteries age, internal resistance tends to increase due to intricate factors such as chemical transformations, mechanical wear, and degradation of electrodes. Dai et al. (2009) defined the SOH as the following [46]:

$$\text{SOH} = \frac{R_{eol} - R_t}{R_{eol} - R_0} \times 100\% \quad (1)$$

where R_0 represents the initial resistance when the battery is new, R_t is the resistance at a time period t , and R_{eol} represents the resistance at the end of life of the battery.

4.2. Capacity Fading

Capacity fading is a phenomenon where the available energy storage capacity gradually diminishes over time and repeated charge–discharge cycles. This degradation occurs due to various electrochemical processes within the battery, leading to a reduced ability to store and deliver the specified amount of energy. Che et al. (2021) used the ratio of current capacity to the rated capacity as the SOH indicator [47,48]. Capacity fading can be represented by the following equation:

$$\text{SOH} = \frac{C_t}{C_0} \times 100\% \quad (2)$$

where C_0 is the rated/nominal capacity when the battery is new, and C_n is the battery capacity at a specific time t .

As argued by Li et al. (2022), capacity fading is a more intuitive health indicator than the internal resistance or voltage [49]. For instance, Nissan's Leaf EVs utilize capacity fade metrics to inform users about battery health. In practice, the influential factors of capacity fade include ambient temperature, current rate, the aging process, and the past aging trajectory [50]. In the automotive industry, the SOH for a new battery is set to be 100% and will gradually decrease. Once the SOH drops below 80%, the EV battery is considered unsuitable for vehicle propulsion and may need to be repurposed for secondary applications, such as energy storage systems [51].

Although the battery capacity and internal resistance have been widely used for SOH indicators, these metrics have very tight assumptions, such as full cycling with constant current [52]. Nevertheless, it is difficult for EV batteries to achieve complete charging and discharging under actual EV working conditions. Thus, the capacity and internal resistance cannot be checked online. Instead, Li et al. (2022) proposed that the constant current drop time (CCDT), constant current drop capacity (CCDC) and maximum constant current drop rate (MCCDR) in constant voltage charging stage can be used as health indicators for an indirect quantification of the battery degradation process [53].

In real-world applications, the choice of SOH indicator depends on the specific operational and environmental context. For example, fleet operators may prioritize metrics that can be monitored online without interrupting normal operations, while individual EV owners may rely on dashboard SOH displays based on capacity fade. Additionally, battery manufacturers are increasingly leveraging hybrid metrics that combine internal resistance, capacity fading, and advanced metrics to provide more comprehensive SOH assessments.

5. Influential Factors for Battery SOH

Existing studies have mainly used inputs such as current, voltage, and temperature. However, other factors, such as weather, traffic conditions, driving patterns, and road conditions, are largely ignored.

5.1. Environmental Factors

Ambient temperature is a critical factor influencing the performance and longevity of EV batteries. Research has indicated that temperature non-uniformity and high temperatures can lead to substantial degradation of battery performance, underscoring the importance of avoiding extreme temperatures to maintain battery health [54]. For example, Chen et al. (2021) compared the capacity degradation of different types of batteries (such as P01 and S43) at different temperatures (such as 25 °C and 10 °C). When the capacity degradation reached 0.4 Ah, the resistance increment of the S43 battery tested at 10 °C was much larger than that of the S30 battery tested at 35 °C, indicating that the temperature significantly affects the removal and insertion speeds of lithium ions, thereby affecting the performance and the battery SOH [55]. Moreover, the thermal characteristics of batteries, such as decay rates, can vary with temperature, with decay rates increasing as temperatures decrease below specific thresholds [56]. Experiments conducted at subzero temperatures have shown the influence of temperature on battery aging, with tests revealing differences in battery performance and capacity fade at various temperature levels [57].

5.2. Operational Factors

- Cycling Behavior and Depth of Discharge;

The cycling behavior of EV batteries, characterized by charge and discharge cycles, significantly influences their lifespan. Aggressive cycling patterns, involving frequent and intensive charge–discharge cycles, can accelerate capacity fade and reduce overall battery lifespan [58]. The depth of discharge (DoD) represents the extent to which a battery's stored energy is utilized during a cycle. Deep discharges, where a significant portion of the battery's capacity is utilized, contribute to stress on electrode materials and can expedite aging processes. Mathematical correlations based on manufacturer's data have highlighted the impact of DoD on battery degradation [59]. As the DoD increases, the temperature rise in batteries also increases, impacting battery chemical performance and lifecycle [60].

- State of Charge;

SOC refers to the remaining charge in a battery, expressed as a percentage of its total capacity. The existing literature underscores the critical role of maintaining optimal SOC levels to mitigate degradation. Shahriari and Farrokhi (2013) developed an online SOH estimation model which is based on the SOC of the battery, indicating the crucial role of SOC in determining the health and remaining life of the battery [61].

- Charge–Discharge Rates;

Charge and discharge rates significantly impact the EV battery SOH. The battery aging rate is closely related to the charging and discharging rates and depth of discharge [62]. Studies have shown that the SOH of batteries is affected by the state of charge (SoC) and the C-rate of charging and discharging, emphasizing the importance of managing these parameters to preserve battery health [63]. For example, Li et al. (2019) conducted a detailed analysis of the battery SOH under different charge and discharge rates (such as 0.5C and 1C). They found that the number of cycles for CS2#33 to reach the end of life (EoL) under a 0.5C discharge rate is 606 times, whereas the number of cycles for CS2#36 to reach EoL under a 1C discharge rate is 667 times. This indicates that the charge and discharge rate has a significant impact on SOH [64]. Optimizing charge and discharge strategies, balancing

performance and longevity, is essential for effective battery management. Additionally, the stress developed inside the battery is related to the rate of charging, highlighting the need to consider charging rates in battery management.

- Driving patterns.

The EV battery SOH is significantly influenced by driving patterns. Factors such as trip distance, acceleration, speed, road grade and driving times can impact the depth of discharge (DOD), current input/output, and heat generated by the battery [65]. For example, Nuhic et al. (2013) investigated the influence of driving patterns (HV1–HV5) on the battery SOH through actual driving cycle experiments. In the experiment, five different driving patterns were employed to conduct cyclic stress tests on the batteries, and each driving pattern encompassed specific speed and acceleration curves to simulate diverse working conditions in real-world driving conditions [66]. Results suggest that the HV1 driving pattern sees the battery SOH faded by about 12%. In contrast, under the HV5 driving pattern, after the same number of cycles, the SOH fade reached about 23%. Integrating usage patterns into EV SOH predictive modeling can enhance battery SOH prediction accuracy. However, many existing studies have often overlooked the impact of driving patterns on battery degradation costs [67].

6. Datasets for Battery SOH Modeling

High-quality data are essential for the development of battery SOH estimation models. Existing studies mainly use three types of data, including offline laboratory testing data, virtual experimental data, and real-world EV battery operational data, as detailed below.

6.1. Offline Laboratory Data

Many prior studies estimated the battery SOH based on laboratory experiments and examined the battery's aging behavior [68,69]. The objectives of these studies are mainly for battery SOH offline prediction. Some studies use internal resistance to predict the SOH. The internal resistance can be obtained via the electrochemical impedance spectroscopy (EIS) tests. For example, Zhang et al. (2020) developed models based on the Gaussian process regression to estimate the battery capacity using the EIS spectrum [70].

Note that the aging mechanism in real-world conditions can be very different from the experimental conditions [71]. However, most of the studies relied on datasets, which fail to represent real-world EV operation conditions [72]. The dynamic real-world driving cycles and weather conditions (e.g., temperatures) are hardly incorporated in commonly used laboratory datasets. Moreover, laboratory experiments, like characterization tests and incremental capacity analysis (ICA), are time-consuming and necessitate costly battery testing facilities. Lastly, some input features can only be obtained using intrusive sensors, which hinders the real-time online applications. Thus, these testing facilities are highly constrained by the growing global demand for high-voltage battery applications [73].

In contrast to the laboratory experiments, Heinrich and Pruckner (2022) presented a new approach for battery SOH estimation based on battery virtual experiments [74]. A long short-term memory (LSTM) neural network model was used to simulate the electric response during capacity testing, incremental capacity analysis, and peak-power testing, which were adapted to battery usage in real-world. Results demonstrated high accuracy for battery SOH estimation in terms of remaining capacity (0.60% MAE) and internal resistance (1.77% MAE). In contrast to the traditional laboratory experiments, the virtual battery experiments are scalable, reducing costs of laboratory efforts and testing facilities.

6.2. Real-World EV Operation Data

Different from the laboratory data, the real-world EV operational data represent a promising alternative and have been increasingly used for battery SOH modeling. Some studies employed data that included variables such as current, voltage, and temperature. These variables can be obtained from the smart battery management system (BMS), which records the battery operating data continuously. Therefore, studies based on real-world EV operational data are very useful for online battery SOH prediction. For example, You et al. (2016) estimated SOH in real-world scenarios, emphasizing the necessity for batteries to accommodate real-world driving patterns. Specifically, the study introduces a data-driven methodology to dynamically monitor SOH, utilizing BMS data, including current, voltage, and temperature, and leveraging their historical distributions. The proposed approach yields exceptionally accurate results in real-world EV driving conditions, with an average error rate of under 2.18% [52]. Zhao et al. (2023) developed a modeling framework for battery SOH prediction using real-world EV operating data. Specifically, the battery aging features were extracted from plenty of historical EV operating data [71].

7. Battery SOH Modeling Approach

The battery SOH modeling approaches can be divided into four categories which is shown in Figure 2: (1) direct measurement method; (2) physics-based; (3) data-driven approach; and (4) hybrid approach. Figure 2 shows the detailed classification of battery SOH estimation models, which will be discussed subsequently.

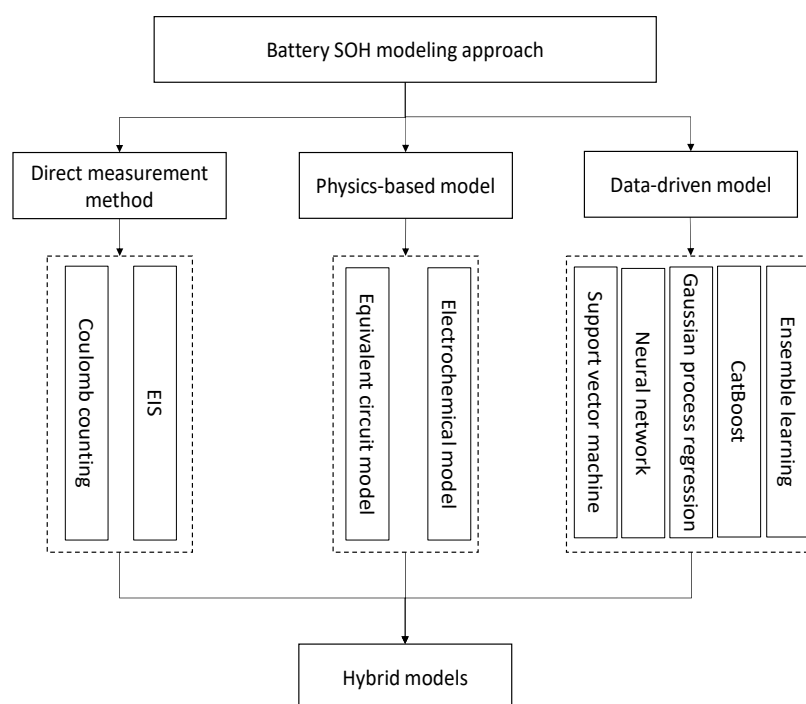


Figure 2. Battery SOH modeling approach.

7.1. Direct Measurement Method

- Coulomb counting (CC);

The coulomb counting (CC) approach estimates the battery capacity directly by tracking the flow of electric charge in or out of the battery during a full cycle. The CC approach is often applied in two situations. First, the CC can be dedicatedly used for battery testing. The major disadvantage is that this approach is time-consuming and labor-intensive. Therefore, the measurements are not immediately applicable as an online approach because

they must be carried out under controlled settings with a constant C-rate and a specified ambient temperature [75]. Moreover, testing the battery to its full capacity (from 0% to 100%) runs the risk of decreasing its lifespan by accelerating degradation.

Second, the CC can be carried out when the system is in operation without a separate test full discharge. Despite being more practical, it incorporates both accumulative measurement errors resulting from the dynamic usage profiles [76]. The CC method is sensitive to errors over time. Therefore, it may require periodic recalibration or correction to maintain accuracy [77].

- Electrochemical impedance spectroscopy (EIS).

The electrochemical impedance (EIS), a real-time, non-invasive measurement, has been used to determine the internal resistance of batteries [70]. For example, Ouyang et al. (2016) examined the battery degradation based on the chemical kinetics [78]. The parameters used in these models are often collected based on the EIS equipment. The EIS enables examining the impedance response of a battery at different frequencies. Changes in impedance spectra over time or usage can provide insights into various degradation mechanisms, such as electrode–electrolyte interface changes, formation of passivation layers, or alterations in internal resistance [79]. Blanke et al. (2005) have demonstrated the connection between the EIS measurements and SOH at different frequencies [80]. However, the EIS equipment is sensitive to changes in electrode interfaces. Therefore, any alterations or contamination at the electrode–electrolyte interface can significantly impact the impedance measurements, potentially leading to misinterpretations of the battery SOH.

7.2. Physics-Based Model

The physics-based models are based on the complex dynamic mechanism of the battery degradation. Thus, this approach provides detailed insights into the battery physical behavior, facilitating accurate simulation of battery degradation mechanisms. However, the physical-based approach also has limitations. Its computational complexity may hinder real-time SOH estimation applications since they often require a large amount of non-linear partial differential equations which are cumbersome to solve [81]. Specifically, the physical-based models can be further divided into two categories: (1) equivalent circuit models, and (2) electrochemical models.

- Equivalent circuit model (ECM);

The equivalent circuit model (ECM) is a simplified physical-based approach to predict battery SOH. The ECM delineates the connection between the internal resistance within a battery and the battery's behavior during the processes of charging and discharging [82]. The equivalent circuit typically consists of idealized circuit elements such as resistors, capacitors, and voltage sources, which are chosen to replicate the observed behavior of the real batteries. The ECM is relatively simple and easy to implement, providing a practical representation of the battery's electrical behavior without the need for extensive computational resources. However, such simplicity of the ECM comes at the cost of SOH prediction accuracy. Moreover, it may struggle to accurately represent highly non-linear and rapid dynamic behaviors exhibited by some batteries under certain operating conditions [83].

Several types of ECMs exist, each offering a different level of complexity, accuracy, and reliability to capture various aspects of battery dynamics. The first type is the Thevenin battery model [84]. It characterizes the battery as an ideal voltage source (Thevenin voltage) in series with an internal resistance (Thevenin resistance). This model assumes that the battery behaves linearly and that its voltage remains constant regardless of the current drawn. This model is particularly useful for analyzing steady-state conditions

and simplifying complex electrical circuits. The second type of ECM is called the nRC model, where “n” represents the number of resistor–capacitor (RC) pairs [85]. The nRC model is a more complex model that considers the transient response and time-dependent characteristics of a battery. The third type of the ECM is the Randle’s circuit model, which is an extension of the basic Thevenin equivalent circuit, introducing additional elements (e.g., capacitance and Warburg impedance) to capture more dynamic behaviors of batteries, particularly in transient responses [86].

For a specific ECM battery model, adaptive filtering and observer algorithms are excellent tools for determining the model parameters which describe the state of the battery. Since the model parameters often change dynamically, the Kalman filter (KF) can be employed to dynamically update model parameters [87]. A variety of improved algorithms based on the Kalman filter have been used to solve the non-linear and high-computational model, including sigma point KF [88], adaptive extended KF [89], extended KF [90], unscented KF [91], and particle filter [92]. In addition to Kalman filtering and its improved algorithms, least square (LS) is another widely used algorithm to identify battery model parameters thanks to its ease of use and low computational processing requirements [93].

- **Electrochemical model.**

The electrochemical models are physics-based, grounded in fundamental electrochemical principles and equations. They consider the complex interactions at the electrode–electrolyte interface, involving chemical reactions and transport phenomena. One of the prominent electrochemical models is the Doyle–Fuller–Newman (DFN) model, which incorporates porous electrode theory, offering spatial resolution to account for variations within electrodes [94,95]. Nevertheless, the electrochemical models are limited in real-time SOH applications due to the complex partial differential equations and high requirements on computational resources.

Similarly to the ECM, the adaptive filter method can also be used to identify model parameters of the electrochemical model for state of health estimation. For example, Zheng et al. (2016) used the trinal proportional–integral (PI) observers with the electrochemical model to estimate the capacity and resistance for lithium-ion batteries [96]. This method was been validated against erroneous starting values, varying aging levels of battery cells, and ambient temperatures. Similarly, extended KF [97] and multi-time-scale observer were also used to estimate battery resistance and capacity [98].

7.3. Data-Driven Approach

Data-driven approaches based on machine learning or deep learning has emerged as a promising avenue for SOH estimation thanks to the greater availability of large-scale battery operation data and advances in computational power and AI algorithms. Unlike the physics-based approach which relies on battery models, the nature of the data-driven approach is to learn the patterns and correlations directly between battery SOH and voltage, current, temperature measurements, and other factors [55,99]. The data-driven approach has demonstrated its strength in capturing complex and non-linear relationships, without resorting to the complex electrochemical process of batteries.

The data-driven approach for battery SOH estimation is often implemented in the following three steps. The first step is the algorithm selection. Various algorithms have been applied to battery SOH prediction, including support vector machines, random forests, decision trees, gradient boosting, etc. The second step is feature selection and engineering, which plays a pivotal role in the process of SOH prediction modeling. Feature selection techniques, such as recursive feature elimination and correlation analysis, are often employed to identify the most influential factors affecting battery SOH. The last step

is training and validation where the models undergo a training phase using historical data, where the algorithm learns to recognize patterns associated with battery SOH. The validation using hold-out datasets assesses the model's generalizability and its ability to make accurate predictions on new, unseen data. Specifically, the existing literature has used a variety of data-driven models for battery SOH estimations, which are summarized as follows. Summary details on the studies based on data-driven approaches are shown in Table A1.

- Support vector machine;

Support vector machine (SVM) is a robust machine learning algorithm for both classification and regression tasks. It operates by identifying the optimal hyperplane that separates data points in a high-dimensional space, maximizing the margin between different classes. SVM is effective in handling both linear and non-linear relationships through the use of kernel functions. Nuhic et al. (2013) estimated battery SOH using SVM at different environment and load conditions [66]. Feng et al. (2019) developed an SVM algorithm to estimate online SOH for Li-ion batteries [100]. The proposed algorithm requires only partial charging curves (e.g., 15 min), enabling fast on-board SOH estimation. Notably, the algorithm was validated against two commercial Li-ion batteries and showed an error lower than 3% error for 95% of all cases for SOH prediction.

Tan et al. (2020) developed a real-time SOH estimation based on the SVM approach, where the SOH is represented by the equivalent internal resistance [101]. The developed method was validated using laboratory testing data, and showed satisfactory prediction accuracy and robustness. Li et al. (2020) used the SVM algorithm to develop a battery degradation model for Li-ion batteries through partial incremental capacity analysis [102]. The study used advanced filter algorithms to smooth incremental capacity curves. Then, the smooth curves were decomposed using a peak fitting technique. The decomposed curves were then used to extract battery health features to train the SVM algorithms. The proposed method showed a mean absolute error (MAE) and root mean square error (RMSE) lower than 2%.

- Neural network;

Neural networks stand out as a prominent tool in the realm of data-driven predictive modeling. It excels in capturing complex, non-linear relationships within datasets, making them well-suited for the dynamic nature of EV battery behavior. Researchers employ neural networks to analyze multifaceted datasets, uncovering hidden patterns that contribute to battery failure. The flexibility of neural networks allows for the integration of various input parameters, providing a versatile approach to battery SOH prediction across different operational conditions. More specifically, the artificial neural networks (ANNs), recurrent neural network (RNN), long short-term memory (LSTM) network, deep convolutional neural networks (DCNNs), and extreme learning machine (ELM) have been used for battery SOH estimation.

Naha et al. (2020) predicted battery SOH based on ANN using only partial charging data during normal usage [103]. The difference between the voltage values and the average temperature during the charging are set as a feature vector for the trained ANN algorithm. The proposed method required a moderate training data size, and achieved less than 1% mean absolute error (MAE) for the test cases. Similar studies using ANN are Rezvani et al. (2011) [104] and Andre et al. (2013) [105]. The recurrent neural network (RNN), which is a recurrent variant of the ANN, has also been used for estimating the SOH of batteries [106,107].

The long short-term memory (LSTM) network (improved variant of standard RNN) has been recently used for SOH estimation. Tan and Zhao (2020) combined the LSTM with

the fully connected (FC) layers based on transfer learning for battery SOH estimation [108]. The proposed method was validated using experimental datasets and demonstrated accurate and robust SOH estimation. Li et al. (2021) proposed the online SOH estimation approach under real-world conditions based on the deep LSTM network [109]. The input feature for the LSTM is the voltage–time sensor data from the partial constant current phase charging curves. The LSTM network demonstrated robustness and achieved a best-case mean absolute percentage error of 0.76%. Similarly, the LSTM was used by Wu et al. (2020) for SOH estimation where the voltage during discharge and charge process and incremental capacity variation were extracted as health features [69].

Shen et al. (2020) estimated the SOH for Li-ion batteries based on deep convolutional neural networks (DCNNs) with ensemble learning and transfer learning [99]. The transfer learning and ensemble learning enable the SOH estimation in applications where only a small training dataset is available. The proposed method is then validated against the dataset from 20 commercial 18650 Li-ion cells. The prediction performance excels compared to the other data-driven methods such as random forest regression and Gaussian process regression.

Pan et al. (2018) developed the extreme learning machine (ELM) algorithm to capture the underlying correlation between the extracted multiple SOH indicators and battery capacity degradation [110]. This proposed approach outperforms the traditional backpropagation neural network with better prediction accuracy and faster speed. Chen et al. (2021) developed a novel metabolic extreme learning machine (MELM) for SOH estimation for different batteries at different usage levels [55]. The results showed the effectiveness of the proposed method in SOH estimation due to the process of degradation feature selection. Similar studies that used the ELM method are Shu et al. (2020) [111], Liu et al. (2020) [112], and Liu and Xu (2020) [113].

- Gaussian process regression (GPR);

Note that the machine learning algorithms above (e.g., SVM, ANN) are often called parametric algorithms, since a specific functional form is assumed to represent the mapping relationship between the SOH output and input features. The underlying coefficients for the assumed functional form are then estimated subsequently based on the training dataset [114]. In contrast, another category of machine learning (e.g., GPR) requires no assumptions about the mapping relationship between the input features and SOH output. They are often referred to as non-parametric algorithms, which will be discussed subsequently.

Yang et al. (2018) proposed a novel GPR model based on charging curves to estimate SOH for Li-ion batteries [115]. Four features, representing the battery aging mechanism from different perspectives, were extracted from the charging curves and were then used as inputs of the GPR models. The proposed method demonstrated high accuracy in SOH estimation based on the battery aging dataset from the NASA data repository. Li et al. (2020) used advanced signal filtering techniques to refine and smooth the initial incremental capacity curves. Then, the feature variables (e.g., the intercept, slope, and peak) were obtained by linear fitting the partial incremental capacity curves [116]. The kernel-modified Gaussian process regression (GPR) was developed based on the inputs of the significant feature variables. The proposed method was verified against four datasets with varying health levels and cycle testing conditions and showed accurate SOH forecast performance.

Sheng et al. (2021) developed weighted GPR for small-sample SOH estimation, considering the fact that large-scale data collection is expensive and time-consuming [117]. Knowledge transfer is employed to address the GPR model's dependence on data. The developed method was validated against experiments with aging data under various work-

ing conditions. Results demonstrated reliable SOH prediction with a training dataset that accounts for only 20% of the total dataset. Deng et al. (2021) used the standard deviations of discharge capacity curves and the difference in discharge capacity between two cycles as the SOH indicators [118]. Then, the SOH prediction performance was compared across linear regression, SVM, relevance vector machine, and GPR under different operation conditions for three battery types. The GPR showed the best prediction performance, with the maximum absolute error and root mean square error lower than 1% and 1.3%, respectively.

Tagade et al. (2020) developed a deep GPR (DGPR) algorithm for the SOH estimation for Li-ion batteries [114]. The proposed DGPR algorithm was validated based on battery aging datasets cycled under various C-rates, temperature, and usage profiles. Results showed the effectiveness of the proposed method in predicting the SOH and RUL, with a coefficient of determinant higher than 0.9 and a mean absolute error below 0.1.

- CatBoost;

The CatBoost algorithm is one of the members of the gradient boosting technique on decision trees. Li et al. (2022) proposed an innovative battery SOH estimation model using Catboost and interval capacity [49]. The study incorporated five input features: start-charging state of charge (SOC), end-charging SOC, mileage, probe temperature, and current. Compared to six other machine learning models, the proposed Catboost-based model achieved the highest accuracy, achieving a mean absolute percentage error below 2.74% and a root mean squared error under 1.12%.

- Ensemble learning;

Instead of relying on one specific machine learning algorithm, some researchers proposed the ensemble learning framework for the battery SOH estimation, which have potential to improve the prediction accuracy of SOH estimation via integrating multiple weak learners. Ensemble learning can thus reduce the risk of choosing a learning algorithm with poor performance by combining prediction results from multiple learning algorithms. For example, Meng et al. (2020) proposed a self-adaptive differential evolution algorithm to integrate multiple weak learners [119]. The proposed method was validated against two LiFePO₄/C batteries which were cycling under a mission profile. Gou et al. (2020) developed a hybrid ensemble data-driven method to forecast the battery SOH for Li-ion batteries, integrating two random learning algorithms for SOH estimation [120]. Similarly, Li et al. (2019) used the ensemble learning method to generate differential data samples, integrating the output of multiple base learners [121]. The proposed method was validated against NASA battery datasets, results suggesting an enhanced prediction accuracy and stability.

- Time-series model.

The data-driven models for battery SOH estimation summarized above are classified based on specific machine learning algorithms. Depending on the input features, these data-driven models can be further classified into three categories: (1) non-linear autoregressive models (NAR); (2) non-linear regressive models with exogenous variables (NRX); and (3) non-linear auto regressive models with exogenous variables (NARX) [44,122].

The NAR approach predicts battery SOH based solely on historical SOH values, which has demonstrated the high efficiency for short-term battery SOH predictions. The NAR approach is often combined with the LSTM algorithm, which has strength in handling time-series data prediction. For example, Li et al. (2019) integrates the empirical model decomposition algorithm and LSTM and Elman neural networks to forecast the remaining useful life of Li-ion batteries [64]. Similar studies using the NAR are Liu et al. (2021) and Li et al. (2021) [123,124]. The NAR approach for SOH estimation often assumes that the battery usage condition remains constant over the course of its lifetime. This assumption, however, is not realistic for EV operation in real-world conditions.

In contrast, the NRX approach just uses the exogenous variables, such as current, voltage, temperature, and internal resistance, to predict the SOH [107,125–127]. For example, Jorge et al. (2023) presented a novel fully exogenous method that does not rely on historical SOH value, but is solely based on aging features that are taken from operating data [122]. More specifically, the evolution of the battery's SOH is predicted based on extracting features from current, voltage and temperature profiles through charging and discharging. An LSTM neural network is trained with a sliding window of cycles as input, and it produces a multi-step-ahead forecast of the battery SOH. The proposed method could account for variations in the vehicle's usage environment or in the users' driving profile.

7.4. Hybrid Models

Recognizing the complementary strengths of physics-based and data-driven models, the hybrid approach has gained prominence in the recent literature. By integrating detailed electrochemical insights from physics-based models with the adaptability and scalability of data-driven approaches, researchers aim to achieve enhanced predictive accuracy. This synergy allows for a more comprehensive understanding of battery behavior, leveraging the strengths of both modeling paradigms to address the inherent challenges posed by the dynamic and multifaceted nature of EV batteries. However, the hybrid models also have limitations, such as the challenges in harmonizing diverse models and the calibration complexity, which hinders its applicability in real-world situations.

Song et al. (2020) developed an SOH estimation approach that integrates the data-driven least-square SVM and model-based unscented-particle-filter [128]. The study used terminal voltage and electric current as the measurement of SOH indicators. The proposed joint modeling approach showed accurate and robust estimation results, with a lower than 4% root mean square error of SOH estimation. Zhang et al. (2020) proposed a prediction method for Li-ion battery SOH by integrating the electrochemical impedance spectroscopy (EIS) and GPR. More than 20,000 EIS spectra of Li-ion batteries under different SOH, states of charge and temperatures were used as inputs for the Gaussian process model [70]. The proposed method highlighted the importance of EIS data in predicting battery SOH and remaining useful life. Park et al. (2017) developed a hybrid modeling framework which combines a recurrent neural network (RNN) and first-principle electrochemical battery model [106]. The RNN captures unmodeled dynamics in electrochemical models. The authors compared the proposed method with reduced electrochemical battery models using simulation. The simulation results showed that the developed hybrid modeling framework outperforms the reduced electrochemical battery models in most cases. Similarly, Lin et al. (2023) combines ECM with data-driven approach [129].

While hybrid models effectively combine the strengths of physics-based and data-driven approaches, they introduce trade-offs between computational demand and prediction accuracy, which must be addressed for practical applications. The detailed electrochemical insights provided by physics-based models often involve solving complex mathematical equations, leading to high computational overhead. When integrated with data-driven methods, such as machine learning models, the computational demand increases further, particularly during the training and inference stages. To address these trade-offs, reduced-order modeling techniques can simplify physics-based components without sacrificing essential insights, while transfer learning has great potential to optimize the training process for data-driven components. Additionally, distributed computing and edge-computing architectures could support the implementation of hybrid models by offloading intensive computations from BMS to cloud platforms. By explicitly addressing these trade-offs, hybrid models can achieve an optimal balance between computational feasibility and predictive accuracy, enhancing their applicability in real-world EV scenarios.

8. Discussion

8.1. Comparison Between Different SOH Modeling Approaches

Table 3 summarizes the different modeling approaches for battery SOH models. Most of them are physical-driven or data-driven. The data-driven approach has advantages in discerning complex relationships between battery operational parameters and SOH, enabling the development of predictive models capable of adapting to diverse battery chemistries and usage scenarios. It is capable of learning complex, non-linear patterns from large datasets, adaptive to changing conditions and non-linear degradation, and suitable for real-time applications. However, it is highly dependent on extensive and diverse datasets for training; robust datasets are crucial for accurate predictions, and lack interpretability compared to physics-based models. In contrast, the physical-based approach simulates the internal thermal behavior of batteries, so that the model results often show higher interpretability compared to the data-driven approach.

Table 3. A comparison of different battery SOH estimation approaches.

Methods		Strength	Limitations
Direct measurement method	Coulomb counting (CC)	Estimates the battery capacity directly	Time-consuming and labor-intensive; Not immediately applicable as an online approach; Sensitive to errors over time;
	Electrochemical impedance spectroscopy (EIS)	Real-time, non-invasive measurement	Data
Physics-based model	Equivalent circuit model (ECM)	Relatively easy to implement, without the need for extensive computational resources	Difficult to accurately represent highly non-linear and rapid dynamic behaviors exhibited by some batteries under certain operating conditions;
	Electrochemical model	Captures the complex interactions at the electrode–electrolyte interface	Limited in real-time SOH applications
Data-driven approach	SVM, LSTM, GPR, Catboost, Ensemble learning	Can capture complex and non-linear relationships, without resorting to the complex electrochemical process of batteries	Has Blackbox nature; Prediction accuracy highly relied on the quality of input dataset; Limited applicability for unseen data
Hybrid models		Allows for a more comprehensive understanding of battery behavior and enhanced predictive accuracy	Hard to be applied in real-world situations due to the challenges in harmonizing diverse models and the calibration complexity

The integration between different modeling approaches represents a promising avenue for future EV battery SOH studies. First, the hybrid models can enhance interpretability and trust of SOH models. One of the major criticisms of the data-driven approaches is their “black-box” nature, which makes them difficult to interpret. By integrating physics-based insights, hybrid models can provide interpretable results, fostering trust for the SOH models. Second, the hybrid approach has potential to address the data scarcity challenge. For example, in the context where operational data are scarce, researchers can use physical-based simulations to generate degradation profiles. The generated profiles can then be fed into the data-driven models, enabling better training for machine learning algorithms. However, the hybrid model is still an emerging field, and it tends to require higher computational resources, which hinders the online applications of SOH monitoring.

Since each approach shows its unique advantages and limitations, no single approach is significantly superior to the others. Instead, these approaches are suitable for different scenarios and research objectives. The direct measurement method is well suited for scenarios requiring quick assessments of battery SOH, such as real-time monitoring in EVs or basic battery management systems with limited computational resources. While the direct measurement method is cost-effective and simple to implement, it often lacks accuracy and generalizability, making it more appropriate for applications where approximate results are sufficient. Physics-based methods, on the other hand, are ideal for detailed investigations into the internal electrochemical processes of batteries. This approach provides high accuracy and interpretability, but their computational complexity and reliance on extensive parameterization limit their scalability for real-time use. Data-driven methods excel in environments with abundant, high-quality datasets, leveraging machine learning algorithms to deliver highly accurate and adaptive SOH predictions. These methods are particularly effective for predictive maintenance and intelligent battery management systems but may struggle in scenarios with limited or noisy data. Lastly, hybrid methods combine the strengths of physics-based and data-driven approaches, offering a balance between accuracy and computational efficiency. These methods are especially beneficial in complex systems that require both interpretability and adaptability, such as integrated fleet management or multi-source energy systems. Each method, therefore, serves distinct objectives and application contexts, highlighting the importance of selecting the suitable approach based on specific requirements and scenarios.

8.2. From Laboratory Test Data to Real-World Big Data

Early studies mainly relied on laboratory testing data due to the limited EV penetration and real-world battery operational data. These studies examined the internal factors related to battery aging. While laboratory datasets have been instrumental in advancing battery SOH estimation, they often fail to capture the variability and complexity of real-world operational conditions. One promising approach to address this limitation is the generation of synthetic datasets using physics-based models. These models can simulate battery behavior under a wide range of scenarios, including extreme temperatures, diverse charge–discharge profiles, and varied usage patterns. By augmenting laboratory datasets with synthetic data, researchers can train more robust and generalizable SOH estimation models.

With the gradual adoption of EVs and lithium-ion batteries, external real-world influential factors such as driving patterns have gradually been incorporated into the battery SOH modeling framework [130]. However, the datasets for SOH model training often represent a snapshot of battery aging status. For data-driven models in particular, the trained models can only be applied to certain battery conditions. With the changing of battery aging status, the model becomes inappropriate and must be retrained with new data, resulting in high training costs. Thus, developing adaptive calibration methods to address changing battery characteristics over time represents a promising avenue for future battery SOH modeling. Moreover, the big data platform with full-life charging data with real-world conditions can be established to provide high-quality data for reliable EV battery SOH estimation.

Lastly, the authors note the caveats of using real-world datasets for SOH modeling. One significant bias stems from region-specific driving conditions. For instance, vehicles in urban areas with frequent stop-and-go traffic show different battery usage patterns compared to those in rural or highway scenarios. Additionally, extreme climates, such as hot regions (e.g., Phoenix, Arizona) or cold regions (e.g., Scandinavia), can disproportionately influence degradation rates, skewing models trained predominantly on such data. Furthermore, data sparsity and imbalance in real-world datasets can exacerbate biases. Regions or conditions with fewer EVs may be under-represented, leading to models that overfit to the dominant scenarios in the dataset. To mitigate these biases, future research should focus on collecting more diverse datasets, incorporating data augmentation techniques, and leveraging synthetic data to fill gaps.

8.3. From the Individual Cell to Battery Pack

Most existing studies focus on the SOH of individual battery cells within the EV's battery pack. In contrast, the SOH of a battery pack must consider the collective performance of all individual cells within it. A large number of battery cells in an EV battery pack may result in inconsistency and impracticality for SOH estimation. Therefore, SOH estimation with high accuracy at the cell level may not hold at the battery pack level. A systematic analysis for battery pack SOH should consider the composition structure of EV battery packs and the inconsistency among the individual battery cells.

8.4. Applicability of Reviewed Models in Semi-Solid or All-Solid-State Batteries

The existing literature mainly studies the characteristics of different semi-solid and all-solid electrolytes for non-large-scale use. There is no accumulation of real vehicle operation data, and it is limited to laboratory research. Therefore, there is less literature on the health status of the batteries [131–136]. Based on the characteristics analysis of semi-solid or all-solid-state batteries, the existing mechanism and data-driven methods are basically suitable for future research on their health status. It is necessary to combine the characteristics of solid-state electrolytes themselves, such as cycle characteristics, life, etc., for parameter re-calibration, and a more applicable and accurate evaluation model needs to be established.

9. Conclusions

This paper conducts a systematic review of studies on EV battery SOH. First, the various indicators for battery SOH are summarized, and the influential factors of battery SOH are introduced, including environmental factors, operational factors, and battery internal structural factors. Moreover, the datasets that were used for estimating battery SOH were examined from the perspective of data sources. This paper shows that the real-world operational factors (e.g., driving patterns and user behavior) are often ignored in existing

datasets for battery SOH estimation. Furthermore, the study reviewed common modeling approaches for battery SOH estimation, with a particular focus on physical-driven and data-driven approaches. Results show the strength and limitations of each modeling approach. Thus, the hybrid approach, which integrates the physical-based and data-driven models, has great potential to improve the prediction accuracy and interpretability of battery SOH model. Lastly, the paper discussed the emerging issues associated with EV battery SOH estimation, including the integration of multiple modeling approaches, collecting battery lifecycle datasets for model development, and moving beyond the individual cells to battery packs.

The findings of this study provide practical implications for both EV manufacturers and battery designers. EV manufacturers can consider adopting hybrid modeling approaches for SOH to implement more accurate and reliable SOH estimation systems, enabling real-time monitoring and proactive maintenance. Additionally, the online SOH indicators, such as constant current drop time (CCDT) and maximum constant current drop rate (MCCDR), can provide actionable insights for real-time battery health monitoring. For battery designers, collecting and utilizing lifecycle data from real-world applications can inform the development of batteries that are more resilient to operational and environmental contexts. Furthermore, moving beyond individual cell-level SOH estimation to pack-level analyses will enable more accurate assessments of overall battery health, improving system-level performance.

For future research, the development of robust datasets is crucial for improving SOH estimation models, particularly for emerging battery chemistries such as solid-state batteries. Current SOH models are heavily reliant on datasets specific to lithium-ion batteries, which may not adequately represent the degradation patterns and operational behaviors of newer chemistries. Future research should prioritize the creation of comprehensive, publicly available lifecycle datasets that include charge–discharge profiles, temperature effects, and aging trajectories under various environmental and operational conditions. Another critical area for future research is the advancement of real-time SOH monitoring systems, particularly at the battery pack level. While most current models focus on individual cell-level analysis, pack-level SOH estimation is essential for assessing the overall health of an EV's battery system. Future research should develop methods to account for cell-to-cell variability, thermal gradients, and inter-cell interactions, which significantly impact pack-level performance.

Author Contributions: All the authors contributed equally in this manuscript. All authors have read and agreed to the published version of the manuscript.

Funding: This research was supported by the Opening Project of Key Laboratory of Operation Safety Technology on Transport Vehicles, Ministry of Transport, PRC [Grant No. KFKT2016-02].

Conflicts of Interest: The authors declare no conflicts of interest. The funders had no role in the design of the study; in the collection, analyses, or interpretation of data; in the writing of the manuscript; or in the decision to publish the results.

Appendix A

Table A1. Summary on the studies based on data-driven approaches.

Literature	Model	Input Features	SOH Indicators	Hardware	Software	Computationally Efficiency	Prediction Accuracy
Nuhic et al. (2013) [66]	Support vector machine (SVM)	SOC, current, cycle number, temperature, time	Internal battery parameters like the actual capacity value	No information	No information	No information	No specific information provided. The validation showed very satisfactory results in the diagnosis of the cells state.
X. Wu et al. (2020) [69]	Long short-term memory (LSTM) network	Five healthy features (HFs) including CC charge duration, CC charge ratio, discharge duration, IC peak value and corresponding voltage	Ratio of current maximum available capacity to the nominal capacity of the battery	No information	No information	Higher computational efficiency, lower RAM occupation, less turning labor and better dominance in solving large-scale parameter optimization	For cell 1, with 30% of total data for training, maximum prediction error is 6.70%; with 60% and 70% of cycle data for training, maximum prediction errors are 1.78% and 1.21%, respectively. For cell 3 with 60% cycle data for training, most estimation errors are lower than 2% and maximum error is 4.55%. When compared with Elman NN, SVM and GPR methods, LSTM shows preferable accuracy with smaller MAE, MSE and RMSE values and higher R ² value
Shen S et al. (2020) [99]	Deep convolutional neural network with ensemble learning and transfer learning (DCNN-ETL)	Voltage, current, and charge capacity curves of a battery cell for one partial charge cycle (matrix with a fixed size of 25 × 3)	Cell capacity	Intel Core processor i7-8700 CPU 3.2 GHz and NVIDIA Titan Xp GPU with 12 GB GDDR5X memory and 64 GByte RAM	Self-devised MATLAB codes running in the MATLAB environment (MATLAB Version 9.6.0.1072779 R2019a, The MathWorks, Inc., Natick, MA, USA)	Training of DCNN-ETL required approximately 10 times more time than training of DCNN-TL on average.	The DCNN-ETL method can produce a higher accuracy and robustness than other data-driven methods (such as random forest regression, Gaussian process regression, DCNN, DCNN-TL, and DCNN-EL) in estimating the capacities of the Li-ion cells. It produced an overall RMSE of 1.503% and had a good generalization capability in battery capacity estimation.
Chen L et al. (2021) [55]	Metabolic extreme learning machine (MELM) framework	Increment of ohmic internal resistance and increment of polarized internal resistance	The initial capacity of the battery and the corrected capacity degradation of the K-th aging cycle	An environmental chamber for temperature control, an ITECH IT6523D charger, an ITECH IT8511 electronic load, tested batteries, and a computer for upper control and data storage	No information	No information	The maximum absolute errors of SOH estimation for different types of batteries were no more than 1.93%. The MELM framework can effectively realize the SOH estimation for different types of batteries with unknown usage levels and shows good accuracy and robustness compared to other data-driven algorithms.

Table A1. Cont.

Literature	Model	Input Features	SOH Indicators	Hardware	Software	Computationally Efficiency	Prediction Accuracy
X. Feng et al. (2019) [100]	Support vector machine (SVM)-based algorithm	Partial charging curves (voltage segments) of length 15 min under constant current charging, SOC (state of charge), etc.	The similarity factor calculated based on comparing the measured partial charging curve with the stored SVM models	Battery Cycler BT-3008, thermal chamber BE-HL-150M3, cells with cathode and graphite anode	MATLAB (used for solving the linear programming problem in establishing the SVM model)	No information	the algorithm can estimate the battery SOH with less than 2% error for 80% of all the cases, and less than 3% error for 95% of all the cases
X. Tan et al. (2020)	Support vector regression (SVR) based on equivalent internal resistance (EIR)	EIR values corresponding to specific states of charge (SoC) within the range of 20–90%	capacity ratio, internal resistance ratio	Battery charge–discharge tester, programmable thermal chamber (KOMEQ) for temperature control, host computer for data recording	No information	No information	the absolute value of the MAE percentage was at most 3%, indicating good robustness
X. Li et al. (2020) [101]	Support vector regression (SVR) based on partial incremental capacity	Battery health features such as peak area, position and height of decomposed incremental capacity (IC) curves after filtering	Battery capacity (calculated by capacity-definition approach using the model output) relative to the rated capacity	Battery test bench including an Arbin BT2000 battery tester, a thermal chamber, and a host computer	Matlab machine learning toolbox	No information	Three battery degradation models were established. For all five tested batteries, the maximum relative errors of the three models are around 3%. The maximum mean absolute error (MAE) and root mean square error (RMSE) are 1.64% and 1.89%, respectively
A. Naha et al. (2020) [103]	Artificial neural network (ANN)	Voltage values and average battery temperature	Battery capacity relative to the maximum capacity of the battery	Thermal chambers for temperature control, batteries with Lithium Cobalt Oxide (LCO) cathode and graphite anode (Type-1 with max capacity 3.0 Ah and Type-2 with max capacity 3.5 Ah)	Python package scikit-learn	No information	Mean absolute error (MAE) is always below 1% and standard deviation (SD) of the absolute error is also below 0.7

Table A1. Cont.

Literature	Model	Input Features	SOH Indicators	Hardware	Software	Computationally Efficiency	Prediction Accuracy
M. Rezvani et al. (2011) [104]	Adaptive neural network (AdNN) and Linear Prediction Error Method (L-PEM)	Battery capacity data over cycles	Battery capacity relative to the rated capacity	Battery test bed for Li-ion batteries at Idaho National Lab, equipment for electrochemical impedance spectroscopy (EIS) measurements	No information	No information	For one-step-ahead battery capacity estimation, AdNN provides a slightly better accuracy with lower mean squared error (MSE) values compared to L-PEM in the given datasets. For battery remaining useful life (RUL) prediction, L-PEM shows higher performance with lower MSE values compared to AdNN, especially when the number of training cycles is relatively low.
D.Andre et al. (2013) [105]	Structured neural network (SNN) and extended Kalman filter (EKF)	Current, temperature, state of charge voltage	Ratio of the actual value to the value at begin of life (BOL)	Vehicle with a hybrid battery (Mercedes Benz S400 Hybrid) containing 35 lithium-ion cells with a nominal capacity of 6.5 Ah each, equipment for measuring current, voltage and temperature	No information	SNN has a lower computational complexity. SNN takes 4.7 s to process an 8700 s drive profile on a commercial computer, while EKF takes 13.5 s for the same profile	For SNN, average voltage deviations of 10–20 mV are in the range of measurement accuracy in vehicles, with maximum deviations of 50–120 mV for high dynamics. For EKF, the ohmic over voltage is estimated accurately, with some incorrect values for high dynamics during the first 4000 s but then converges to correct values
Park, Saehong et al. (2017) [106]	A hybrid model combining Single Particle Model (SPM) and Elman Network (a type of recurrent neural network, RNN)	Input current	Voltage related to battery	No information	No information	No information	hybrid models (SPM + Elman 1 and SPM + Elman 2) show lower RMSE compared to SPM and SPMe in many cases
G.-W. You et al. (2017) [107]	Recurrent neural network (RNN)-based models	Current and voltage	Capacity	A cyler (simulation equipment for charging and discharging batteries), battery cells of type 18650 with a capacity of approximately 3.1 Ah at room temperature, chambers for temperature control	Mini-batch stochastic gradient descent (SGD) method, backpropagation through time (BPTT) for RNN optimization, ADADELTA for adjusting the learning step	No information	The average error is lower than 0.0765 Ah (2.46%) over all experiments. Different model variations and approaches are evaluated using root mean squared error (RMSE).

Table A1. Cont.

Literature	Model	Input Features	SOH Indicators	Hardware	Software	Computationally Efficiency	Prediction Accuracy
Y. Tan and G. Zhao (2020) [108]	Long short-term memory (LSTM) network combined with fully connected (FC) layers as the base model, with a feature expression scoring (FES) rule for task selection and transfer learning strategies	Nine geometric features extracted from the voltage curve during the constant current (CC) charging phase of the battery	Battery capacity relative to the rated capacity	Lenovo ThinkCentre M8500t computer equipped with Intel Xeon W-2123 (3.60 GHz) CPU, 32 GB RAM, and Microsoft Windows 10 ultimate operating system	Keras 2.1.5 upon Tensorflow 1.2.1 backend in Python 3.6	The processing efficiency of the transfer learning has increased by about 35% for B5 prediction task, and about 43% for CALCE prediction tasks, compared to the LSTM-FC. Even for B6 task with rebuilding the FC layers, the efficiency is still improved by 10% than the base model.	Root mean squared error (RMSE) and standard deviation of error (SDE) were used as performance metrics. The LSTM-FC-TL model achieved better RMSE and SDE values.
W. Li et al. (2021) [109]	Long short-term memory recurrent neural network (LSTM-RNN) architecture with bidirectional wrapper	Voltage and time samples from the partial constant current (CC) phase charging curve of the cell	Remaining cell capacity relative to the nominal capacity	Cloud server for model training, NVIDIA Jetson Nano for processor-in-the-loop (PIL) validation	Python 3 environment with TensorFlow 1.14 as the backend and the Keras deep learning library for layer creation	Fast computation times, as measured in an embedded device at around 2 s per sample charging curve	Best-case mean absolute percentage error (MAPE) of 0.76% and worst-case MAPE of 2.08% for normal sensor input; MAPE of 0.77% (best-case) and 2.09% (worst-case) for noisy sensor input; MAPE of 1.04% (best-case) and 4.26% (worst-case) for incomplete sensor input
Pan et al. (2018) [110]	Extreme learning machine (ELM)	Ohmic internal resistance, polarized internal resistance	Battery capacity	Intel Core i3-4150 CPU, 8 GB RAM, Windows 7	MATLAB R2014a	High efficiency due to ELM; time for estimation much faster than BP neural network	MAE: 0.0238 Ah, RMSE: 0.0280 Ah; estimation error for capacity degradation < 0.06 Ah; max error for SOH estimation < 2.5%; better performance and robustness than BP neural network.
Shu et al. (2020) [111]	Fixed-size LS-SVM with Mixed Kernel	Short-term charging data	Capacity degradation (SOH error < 2%)	No information	No information	Efficient due to Extreme Learning Machine (ELM); computational overhead optimized via SA algorithm.	MAE, ME, and RMSE lower than 0.35%, 1.9%, and 0.46%, respectively; maximum error < 2%. Accurate and robust estimation.
Wei Liu et al. (2020) [112]	Hierarchical ensemble model with extreme learning machine (ELM)	Voltage variance during equal time interval (VVETI) derived from small segments of charge–discharge curves	Remaining capacity (current capacity/nominal capacity)	No information	No information	High computational speed with better generalization performance; faster than traditional machine learning algorithms.	RMSE below 0.5% under appropriate voltage ranges across datasets; high accuracy compared to other models.

Table A1. Cont.

Literature	Model	Input Features	SOH Indicators	Hardware	Software	Computationally Efficiency	Prediction Accuracy
Wei Liu et al. (2020) [113]	Extreme learning machine (ELM) with Savitzky–Golay filter	Voltage sequences and discharge rates	Current capacity/Initial capacity	No information	No information	Much faster than traditional iterative algorithms; optimized for high efficiency.	Average RMSE: 1.23%; robust across different discharge rates; absolute errors < 5% in most cases.
Tagade et al. (2020) [114]	Deep Gaussian process regression (DGPR)	Voltage, current, and temperature time-series data from discharge profiles	Battery capacity, end of life (EOL), degradation mode	No information	TensorFlow	High computational efficiency due to deep learning architecture; training with Adam optimizer	$R^2 > 0.9$ and Mean Absolute Error < 0.1 across test cases.
Yang et al. (2018) [115]	Gaussian process regression (GPR) with Double Squared Exponential Kernel	Four features extracted from charging curves (time of CC mode, time of CV mode, slopes at different points)	Capacity ratio	No information	No information	Training phase: high reliability; Testing phase: 95% confidence range	RMSE < 2% for static profiles, RMSE~6% for dynamic profiles.
Li et al. (2020) [116]	Multi-time-scale framework combining GPR-based model and non-linear regression	Incremental capacity (IC) features: slope, intercept, peaks	Short-term SOH and long-term RUL	NASA datasets, no specific hardware mentioned	No information	Efficient for short-term and long-term estimation	MAE < 0.0040 Ah (SOH), <12 cycles (RUL); RMSE < 0.0061 Ah (SOH), <15 cycles (RUL); high robustness.
Sheng et al. (2021) [117]	Weighted Gaussian process regression (WGPR)	Incremental capacity curve, charging–discharging time, resistance (100% SOC)	Battery capacity degradation	Intel Core i7-7800k, 16 GB RAM	MATLAB 2016a for WGPR, Python 3.6 (Keras, TensorFlow for comparison)	Efficient in cross-battery tasks; adapts well to small sample sizes	RMSE: 1.09–7.76, MAE: 0.85–4.57; significant improvement in cross-battery SOH tasks, especially under concept drift scenarios.
Deng et al. (2021) [118]	Gaussian process regression (GPR), SVM, RVM, LR	Standard deviation of $Q(V)$, $\Delta Q(V)$	Battery capacity	No information	No information	Computationally efficient for all conditions	GPR achieved MAE < 1% and RMSE < 1.3% across cases.
Jinhao Meng et al. (2020) [119]	Optimized ensemble learning with SVR and self-adaptive differential evolution (SaDE)	Voltage responses from short-term current pulse tests	Battery capacity and internal resistance	FuelCon battery test station, LiFePO ₄ /C batteries	No information	Fitness converges within 30 generations; optimized weight distribution across learners	MSE: 0.0017 for Cell 1 and 0.0028 for Cell 2; outperforms SVR-based weak learners and other ensemble approaches.
Gou, Xu and Feng (2020) [120]	State-of-health (SOH) Remaining-useful-life (RUL) Non-linear autoregressive (NAR) structure	SOH, Constant current (CC)	Features extracted from the charging voltage signal DECVD optimized through Pearson correlation coefficient selection	Do not require any other additional hardware or system downtime.	No information	The learning speed of ELM and RVFL is thousands of times faster than traditional algorithms such as Support Vector Machines (SVM), and requires less computational memory.	The non-linear autoregressive (NAR) structure is designed to reduce the RUL prediction error of each learning model. Bootstrap-based uncertainty management method is designed.

Table A1. Cont.

Literature	Model	Input Features	SOH Indicators	Hardware	Software	Computationally Efficiency	Prediction Accuracy
Jorge et al. (2023) [122]	State of health (SOH) prediction model for lithium-ion batteries based on time-series feature extraction and long short-term memory (LSTM) network. (SOH-window-XLSTM)	Time-series features (TSFs) and historical features (HFs) extracted from current, voltage, and temperature curves	Evolution of battery health status. Time domain features (such as total energy, area under the curve), statistical features (such as mean, root mean square), and spectral features (such as fundamental frequency, maximum frequency)	No information	No information	No information	Multiple models were trained on the MIT dataset to predict SOH after 25 to 400 cycles. For short-term forecasting (such as after 25 cycles), the optimal RMSPE of the model can reach 1.14%. Even in the long-term prediction after up to 400 cycles, RMSPE remains at a relatively low level of 3.14%. The prediction accuracy of the model on the NASA dataset is still considerable, with the optimal RMSPE reaching 1.9%.
Li et al. (2021) [64]	Sequence-to-Sequence (S2S) neural network architecture based on long short-term memory (LSTM)	Remaining Capacity Array	Remaining capacity, Capacity Degradation Curve, Endpoint (EOL80, EOL65) and Knee point, MAPE	NVIDIA Jetson Nano	Sequence-to-Sequence, S2S	One-shot Prediction Compared with the battery degradation prediction model based on iterative methods, this method improves the calculation speed by nearly 15 times	The model can make accurate predictions using only 100 cycles of data (approximately 7% of the total battery life). In the best case, the initial prediction error is 1.8% and the average error is 1.2%; In the worst case, the initial prediction error is 4.2% and the average error is 3.1%. The median absolute percentage error (MAPE) for predicting the entire lifecycle is 1.1% in the best case and 3.1% in the worst case. The prediction error of EOL80 is 0.9% and 2.7% in the best and worst cases, respectively; The prediction error of EOL65 is 1.1% and 3.4% in the best and worst cases, respectively; The prediction error of inflection point is 0.7% and 2.5% in the best and worst cases, respectively. Under noisy conditions, the initial prediction errors for the best and worst cases are 0.7% and 4.9%, respectively, with average errors of 1.5% and 4.1%, and median errors of 1.3% and 3.6%, respectively.

Table A1. Cont.

Literature	Model	Input Features	SOH Indicators	Hardware	Software	Computationally Efficiency	Prediction Accuracy
Song et al. (2020) [128]	Data-driven LS-SVM (least-square support vector machine) model Model-driven UPF (unscented particle filter) model	Terminal voltage Electric current	Battery performance	No information	No information	No specific The article chose east-squares support vector machine (LS-SVM) as the data-driven algorithm and pointed out that compared with standard support vector machine (SVM), LS-SVM significantly improves the computational speed by replacing inequality constraints with equality constraints, transforming quadratic programming problems into solutions of linear KKT systems	The maximum SOH estimation error is less than 2%. The root mean square error (RMSE) is less than 4%. For the three battery cells tested, the maximum mean absolute error (MAE) of SOH estimation is 0.018, the maximum RMSE is 0.0038, and the maximum standard deviation (SD) is 0.019. The maximum error in SOC estimation during the battery lifecycle is less than 0.005. For the three battery cells tested, the estimated MAE of SOC is between 0.0024 and 0.0038, RMSE is between 0.0026 and 0.0034, and SD is between 0.041 and 0.043.
Hong, J et al. (2021) [130]	variable-length-input long short-term memory (LSTM) neural network	Pack voltage, brake pedal stroke value, motor speed, vehicle speed, current, state of charge (SOC), humidity, air temperature, mileage, average probe temperature, maximum probe temperature, standard deviation (SD) of probe temperatures, SD of cell voltages	Internal resistance (IR) before charging (IRBC)	The system was trained and tested using real-world data collected from an electric taxi equipped with a high-energy lithium-ion phosphate battery pack.	Adam optimizer	Allowing variable-length input sequences, reducing preprocessing efforts. Using optimized hyperparameters: batch size of 8 and 50 training epochs, which balanced training speed and accuracy.	Mean Relative Error (MRE): 0.0546% Maximum Absolute Error (MAX): 0.1323% Root Mean Square Error (RMSE): 0.232% Mean Squared Error (MSE): 0.0538%

References

1. Transportation Emissions Worldwide—Statistics & Facts. Available online: <https://www.statista.com/topics/7476/transportation-emissions-worldwide/#editorsPicks> (accessed on 6 September 2024).
2. IEA. *Global EV Outlook 2023*; IEA: Paris, France, 2023.
3. Hashimi, S.R.; Mahajan, A.M.; Farhad, S. Online estimation of battery model parameters and state of health in electric and hybrid aircraft application. *Energy* **2021**, *229*, 120699. [\[CrossRef\]](#)
4. Opitz, A.; Badami, P.; Shen, L.; Vignarooban, K.; Kannan, A.M. Can Li-Ion batteries be the panacea for automotive applications? *Renew. Sustain. Energy Rev.* **2017**, *68*, 685e692. [\[CrossRef\]](#)
5. Hu, X.S.; Xu, L.; Lin, X.K.; Pecht, M. Battery lifetime prognostics. *Joule* **2020**, *4*, 310–346. [\[CrossRef\]](#)
6. Berecibar, M.; Gandiaga, I.; Villarreal, I.; Omar, N.; Van Mierlo, J.; Van den Bossche, P. Critical review of state of health estimation methods of Li-ion batteries for real applications. *Renew. Sustain. Energy Rev.* **2016**, *56*, 572–587. [\[CrossRef\]](#)
7. Xiong, R.; Li, L.; Tian, J. Towards a smarter battery management system: A critical review on battery state of health monitoring methods. *J. Power Sources* **2018**, *405*, 18–29. [\[CrossRef\]](#)
8. Ungurean, L.; Cârstoiu, G.; Micea, M.V.; Groza, V. Battery state of health estimation: A structured review of models, methods and commercial devices. *Int. J. Energy Res.* **2017**, *41*, 151–181. [\[CrossRef\]](#)
9. Lipu, M.H.; Hannan, M.A.; Hussain, A.; Hoque, M.M.; Ker, P.J.; Saad, M.H.M.; Ayob, A. A review of state of health and remaining useful life estimation methods for lithium-ion battery in electric vehicles: Challenges and recommendations. *J. Clean. Prod.* **2018**, *205*, 115–133. [\[CrossRef\]](#)
10. Li, Y.; Liu, K.; Foley, A.M.; Zülke, A.; Berecibar, M.; Nanini-Maury, E.; Van Mierlo, J.; Hoster, H.E. Data-driven health estimation and lifetime prediction of lithium-ion batteries: A review. *Renew. Sustain. Energy Rev.* **2019**, *113*, 109254. [\[CrossRef\]](#)
11. Hu, X.; Zou, C.; Zhang, C.; Li, Y. Technological Developments in Batteries: A Survey of Principal Roles, Types, and Management Needs. *IEEE Power Energy Mag.* **2017**, *15*, 20–31. [\[CrossRef\]](#)
12. Cao, X.W.; Ma, C.; Luo, L.; Chen, L.; Cheng, H.; Orenstein, R.S.; Zhang, X.W. Nanofiber Materials for Lithium-Ion Batteries. *Adv. Fiber Mater.* **2023**, *5*, 1141–1197.
13. Whittingham, M.S.; Gamble, F.R. The lithium intercalates of the transition metal dichalcogenides. *Mater. Res. Bull.* **1975**, *10*, 363–371. [\[CrossRef\]](#)
14. MacNeil, D.D.; Dahn, J.R. The Reaction of Charged Cathodes with Nonaqueous Solvents and Electrolytes: I. $\text{Li}_{0.5}\text{CoO}_2$. *J. Electrochem. Soc.* **2001**, *148*, A1205.
15. Ohzuku, T. Electrochemistry and Structural Chemistry of $\text{LiNiO}_2(\text{R3m})$ for 4 Volt Secondary Lithium Cells. *J. Electrochem. Soc.* **1993**, *147*, 1862. [\[CrossRef\]](#)
16. Li, W.; Reimers, J.N.; Dahn, J.R. In situ X-ray diffraction and electrochemical studies of $\text{Li}_{1-x}\text{NiO}_2$. *Solid. State Ion* **1993**, *67*, 123–130. [\[CrossRef\]](#)
17. Armstrong, A.R.; Bruce, P.G. Synthesis of layered LiMnO_2 as an electrode for rechargeable lithium batteries. *Nature* **1996**, *381*, 499–500. [\[CrossRef\]](#)
18. Bruce, P.G.; Armstrong, A.R.; Gitzendanner, R.L. New intercalation compounds for lithium batteries: Layered LiMnO_2 . *J. Mater. Chem.* **1999**, *9*, 193–198. [\[CrossRef\]](#)
19. Ohzuku, T.; Makimura, Y. Layered lithium insertion material of $\text{LiNi}_{1/2}\text{Mn}_{1/2}\text{O}_2$: A possible alternative to LiCoO_2 for advanced lithium-ion batteries. *Chem. Lett.* **2001**, *30*, 744–745. [\[CrossRef\]](#)
20. Chen, C.; Xu, X.; Chen, S.; Zheng, B.; Shui, M.; Xu, L.; Zheng, W.; Shu, J.; Cheng, L.; Feng, L.; et al. The preparation and characterization of iron fluorides polymorphs $\text{FeF}_3 \cdot 0.33\text{H}_2\text{O}$ and $\beta\text{-FeF}_3 \cdot 3\text{H}_2\text{O}$ as cathode materials for lithium-ion batteries. *Mater. Res. Bull.* **2015**, *64*, 187–193. [\[CrossRef\]](#)
21. Ma, Y.; Huang, A.; Zhou, H.; Ji, S.; Zhang, S.; Li, R.; Yao, H.; Cao, X.; Jin, P. Template-free formation of various V_2O_5 hierarchical structures as cathode materials for lithium-ion batteries. *J. Mater. Chem. A* **2017**, *5*, 6522–6531. [\[CrossRef\]](#)
22. Li, H.; Zhou, H. Enhancing the performances of Li-ion batteries by carbon-coating: Present and future. *Chem. Commun.* **2012**, *48*, 1201–1217. [\[CrossRef\]](#)
23. Dileo, R.A.; Castiglia, A.; Ganter, M.J.; Rogers, R.E.; Cress, C.D.; Raffaele, R.P.; Landi, B.J. Enhanced capacity and rate capability of carbon nanotube based anodes with titanium contacts for lithium ion batteries. *ACS Nano* **2010**, *10*, 6121–6131. [\[CrossRef\]](#)
24. Bindumadhavan, K.; Srivastava, S.K.; Mahanty, S. MoS_2 -MWCNT hybrids as a superior anode in lithium-ion batteries. *Chem. Commun.* **2013**, *49*, 1823–1825. [\[CrossRef\]](#) [\[PubMed\]](#)
25. Stankovich, S.; Dikin, D.A.; Dommett, G.H.B.; Kohlhaas, K.M.; Zimney, E.J.; Stach, E.A.; Piner, R.D.; Nguyen, S.B.T.; Ruoff, R.S. Graphene-based composite materials. *Nature* **2006**, *442*, 282–286. [\[CrossRef\]](#) [\[PubMed\]](#)
26. Liang, M.; Zhi, L. Graphene-based electrode materials for rechargeable lithium batteries. *J. Mater. Chem.* **2009**, *19*, 5871–5878. [\[CrossRef\]](#)
27. Brownson, D.A.C.; Kampouris, D.K.; Banks, C.E. An overview of graphene in energy production and storage applications. *J. Power Sources* **2011**, *196*, 4873–4885. [\[CrossRef\]](#)

28. Liu, Y.; Artyukhov, V.I.; Liu, M.; Harutyunyan, A.R.; Yakobson, B.I. Feasibility of lithium storage on graphene and its derivatives. *J. Phys. Chem. Lett.* **2013**, *4*, 1737–1742. [\[CrossRef\]](#)
29. Mahmoud, A.; Amarilla, J.M.; Lasri, K.; Saadoun, I. Influence of the synthesis method on the electrochemical properties of the Li₄Ti₅O₁₂ spinel in Li-half and Li-ion full-cells. A systematic comparison. *Electrochim. Acta* **2013**, *93*, 163–172. [\[CrossRef\]](#)
30. Moretti, A.; Kim, G.T.; Bresser, D.; Renger, K.; Paillard, E.; Marassi, R.; Winter, M.; Passerini, S. Investigation of different binding agents for nanocrystalline anatase TiO₂ anodes and its application in a novel, green lithium-ion battery. *J. Power Sources* **2013**, *221*, 419–426. [\[CrossRef\]](#)
31. Cavers, H.; Molaiyan, P.; Abdollahifar, M.; Lassi, U.; Kwade, A. Perspectives on Improving the Safety and Sustainability of High Voltage Lithium-Ion Batteries Through the Electrolyte and Separator Region. *Adv. Energy Mater.* **2022**, *12*, 2200147. [\[CrossRef\]](#)
32. Adenusi, H.; Chass, G.A.; Passerini, S.; Tian, K.V.; Chen, G. Lithium batteries and the solid electrolyte interphase(SEI)—Progress and outlook. *Adv. Energy Mater.* **2023**, *13*, 2203307. [\[CrossRef\]](#)
33. Ko, J.M.; Min, B.G.; Kim, D.W.; Ryu, K.S.; Kim, K.M.; Lee, Y.G.; Chang, S.H. Thin-film type Li-ion battery, using a polyethylene separator grafted with glycidyl methacrylate. *Electrochim. Acta* **2004**, *50*, 367–370. [\[CrossRef\]](#)
34. Deng, D. Li-ion batteries: Basics, progress, and challenges. *Energy Sci. Eng.* **2015**, *3*, 385–418. [\[CrossRef\]](#)
35. Chandler, J.H.; Head, D.A.; Hubbard, M.E.; Neville, A.; Jayne, D.G.; Culmer, P.R. The impact of electrode resistance on the biogalvanic characterization technique. *Physiol. Meas.* **2017**, *38*, 101–115. [\[CrossRef\]](#) [\[PubMed\]](#)
36. Evans, J.W. Introduction and the Significance of Electrometallurgy. *Ref. Module Mater. Sci. Mater. Eng.* **2016**, 1–13. [\[CrossRef\]](#)
37. da Silva, S.P.; da Silva, P.R.C. Alexandre Urbano e Jair Scarminio, Analysis of A Commercial Portable Lithium-Ion Battery Under Low Current Charge–discharge Cycles. *Quim. Nova* **2016**, *39*, 901–905.
38. Roselin, L.S.; Juang, R.S.; Hsieh, C.T.; Sagadevan, S.; Umar, A.; Selvin, R.; Hegazy, H.H. Recent Advances and Perspectives of Carbon-Based Nanostructures as Anode Materials for Li-ion Batteries. *Materials* **2019**, *12*, 1229. [\[CrossRef\]](#) [\[PubMed\]](#)
39. Malik, R.; Burch, D.; Bazant, M.; Ceder, G. Particle size dependence of the ionic diffusivity. *Nano Lett.* **2010**, *10*, 4123–4127. [\[CrossRef\]](#) [\[PubMed\]](#)
40. Poizot, P.; Laruelle, S.; Grugeon, S.; Dupont, L.; Tarascon, J.M. Nanosized transition-metal oxides as negative-electrode materials for lithium-ion batteries. *Nature* **2000**, *407*, 496–499. [\[CrossRef\]](#) [\[PubMed\]](#)
41. Bekaert, E.; Balaya, P.; Murugavel, S.; Maier, J.; Ménétrier, M. 6Li MAS NMR investigation of electrochemical lithiation of RuO₂: Evidence for an interfacial storage mechanism. *Chem. Mater.* **2009**, *21*, 856–861. [\[CrossRef\]](#)
42. Wei, Z.B.; Zhao, J.Y.; Ji, D.X.; Tseng, K.J. A multi-timescale estimator for battery state of charge and capacity dual estimation based on an online identified model. *Appl. Energy* **2017**, *204*, 1264–1274. [\[CrossRef\]](#)
43. Hu, X.S.; Jiang, J.C.; Cao, D.P.; Egardt, B. Battery health prognosis for electric vehicles using sample entropy and sparse bayesian predictive modeling. *IEEE Trans. Ind. Electron.* **2016**, *63*, 2645–2656. [\[CrossRef\]](#)
44. Khaleghi, S.; Karimi, D.; Beheshti, S.H.; Hosen, M.S.; Behi, H.; Berecibar, M.; Van Mierlo, J. Online health diagnosis of lithium-ion batteries based on nonlinear autoregressive neural network. *Appl. Energy* **2021**, *282*, 116159. [\[CrossRef\]](#)
45. Richardson, R.R.; Birkel, C.R.; Osborne, M.A.; Howey, D.A. Gaussian process regression for in situ capacity estimation of lithium-ion batteries. *IEEE Trans. Ind. Inf.* **2018**, *15*, 127–138. [\[CrossRef\]](#)
46. Dai, H.; Wei, X.; Sun, Z. A new SOH prediction concept for the power lithium-ion battery used on HEVs. In Proceedings of the Vehicle Power and Propulsion Conference, VPPC'09, Dearborn, MI, USA, 7–11 September 2009; pp. 1649–1653.
47. Che, Y.H.; Deng, Z.W.; Lin, X.K.; Hu, L.; Hu, X.S. Predictive Battery health management with transfer learning and online model correction. *IEEE Trans. Veh. Technol.* **2021**, *70*, 1269–1277. [\[CrossRef\]](#)
48. Liu, K.L.; Li, Y.; Hu, X.S.; Lucu, M.; Widanage, W.D. Gaussian process regression with automatic relevance determination kernel for calendar aging prediction of lithium-ion batteries. *IEEE Trans. Ind. Inf.* **2020**, *16*, 3767–3777. [\[CrossRef\]](#)
49. Li, R.; Hong, J.; Zhang, H.; Chen, X. Data-driven battery state of health estimation based on interval capacity for real-world electric vehicles. *Energy* **2022**, *257*, 124771. [\[CrossRef\]](#)
50. Rezvanizani, S.M.; Liu, Z.C.; Chen, Y.; Lee, J. Review and recent advances in battery health monitoring and prognostics technologies for electric vehicle (EV) safety and mobility. *J. Power Sources* **2014**, *256*, 110–124. [\[CrossRef\]](#)
51. Severson, K.A.; Attia, P.M.; Jin, N.; Perkins, N.; Jiang, B.B.; Yang, Z.; Chen, M.H.; Aykol, M.; Herring, P.K.; Fraggadakis, D.; et al. Data-driven prediction of battery cycle life before capacity degradation. *Nat. Energy* **2019**, *4*, 383–391. [\[CrossRef\]](#)
52. You, G.W.; Park, S.; Oh, D. Real-time state-of-health estimation for electric vehicle batteries: A data-driven approach. *Appl. Energy* **2016**, *176*, 92–103. [\[CrossRef\]](#)
53. Li, N.; He, F.; Ma, W.; Wang, R.; Jiang, L.; Zhang, X. An indirect state-of-health estimation method based on improved genetic and back propagation for online lithium-ion battery used in electric vehicles. *IEEE Trans. Veh. Technol.* **2022**, *71*, 12682–12690. [\[CrossRef\]](#)
54. Wang, Y. Research on the thermal management system of battery for electric vehicles. *J. Phys. Conf. Ser.* **2022**, *2390*, 012076. [\[CrossRef\]](#)

55. Chen, L.; Wang, H.M.; Liu, B.H.; Wang, Y.J.; Ding, Y.H.; Pan, H.H. Battery state-of-health estimation based on a metabolic extreme learning machine combining degradation state model and error compensation. *Energy* **2021**, *215*, 119078. [[CrossRef](#)]
56. Wang, Y.; Ma, C. Cfd-based numerical analysis of the thermal characteristics of an electric vehicle power battery. *Fluid. Dyn. Mater. Process.* **2022**, *18*, 159–171. [[CrossRef](#)]
57. Jaguemont, J.; Boulon, L.; Venet, P.; Dubé, Y.; Sari, A. Lithium-ion battery aging experiments at subzero temperatures and model development for capacity fade estimation. *IEEE Trans. Veh. Technol.* **2016**, *65*, 4328–4343. [[CrossRef](#)]
58. Jia, B.; Guan, Y.; Wu, L. A state of health estimation framework for lithium-ion batteries using transfer components analysis. *Energies* **2019**, *12*, 2524. [[CrossRef](#)]
59. Zhou, C.; Qian, K.; Allan, M.; Zhou, W. Modeling of the cost of ev battery wear due to v2g application in power systems. *IEEE Trans. Energy Convers.* **2011**, *26*, 1041–1050. [[CrossRef](#)]
60. Lai, Y.; Du, S.; Ai, L.; Ai, L.; Cheng, Y.; Tang, Y.; Jia, M. Insight into heat generation of lithium ion batteries based on the electrochemical-thermal model at high discharge rates. *Int. J. Hydrogen Energy* **2015**, *40*, 13039–13049. [[CrossRef](#)]
61. Shahriari, M.; Farrokhi, M. Online state-of-health estimation of vrla batteries using state of charge. *IEEE Trans. Ind. Electron.* **2013**, *60*, 191–202. [[CrossRef](#)]
62. Chen, Z.; Sun, M.; Shu, X.; Xiao, R.; Shen, J. Online state of health estimation for lithium-ion batteries based on support vector machine. *Appl. Sci.* **2018**, *8*, 925. [[CrossRef](#)]
63. Lee, J.; Kim, J.; Ryu, K.; Won, C. An energy storage system's operational management and control method considering a battery system. *Electronics* **2020**, *9*, 356. [[CrossRef](#)]
64. Li, X.; Zhang, L.; Wang, Z.; Dong, P. Remaining useful life prediction for lithium-ion batteries based on a hybrid model combining the long short-term memory and elman neural networks. *J. Energy Storage* **2019**, *21*, 510–518. [[CrossRef](#)]
65. Wang, D.; Coignard, J.; Zeng, T.; Zhang, C.; Saxena, S. Quantifying electric vehicle battery degradation from driving vs. vehicle-to-grid services. *J. Power Sources* **2016**, *332*, 193–203. [[CrossRef](#)]
66. Nuhic, A.; Terzimehic, T.; Soczka-Guth, T.; Buchholz, M.; Dietmayer, K. Health diagnosis and remaining useful life prognostics of lithium-ion batteries using data-driven methods. *J. Power Sources* **2013**, *239*, 680–688. [[CrossRef](#)]
67. Ginigeme, K.; Wang, Z. Distributed optimal vehicle-to-grid approaches with consideration of battery degradation cost under real-time pricing. *IEEE Access* **2020**, *8*, 5225–5235. [[CrossRef](#)]
68. Yuan, H.; Dung, L. Offline state-of-health estimation for high-power lithium-ion batteries using three-point impedance extraction method. *IEEE Trans. Veh. Technol.* **2017**, *66*, 2019–2032. [[CrossRef](#)]
69. Wu, Y.; Xue, Q.; Shen, J.; Lei, Z.; Chen, Z.; Liu, Y. State of health estimation for lithium-ion batteries based on healthy features and long short-term memory. *IEEE Access* **2020**, *8*, 28533–28547. [[CrossRef](#)]
70. Zhang, Y.; Tang, Q.; Zhang, Y.; Wang, J.; Stimming, U.; Lee, A.A. Identifying degradation patterns of lithium ion batteries from impedance spectroscopy using machine learning. *Nat. Commun.* **2020**, *11*, 1706. [[CrossRef](#)] [[PubMed](#)]
71. Zhao, X.; Hu, J.; Hu, G.; Qiu, H. A state of health estimation framework based on real-world electric vehicles operating data. *J. Energy Storage* **2023**, *63*, 107031. [[CrossRef](#)]
72. Vidal, C.; Malysz, P.; Kollmeyer, P.; Emadi, A. Machine learning applied to electrified vehicle battery state of charge and state of health estimation: State-of-the-art. *IEEE Access* **2020**, *8*, 52796–52814. [[CrossRef](#)]
73. Rumberg, B.; Schwarzkopf, K.; Epding, B.; Stradtman, I.; Kwade, A. Understanding the different aging trends of usable capacity and mobile Li capacity in Li-ion cells. *J. Power Sources* **2019**, *22*, 336–344. [[CrossRef](#)]
74. Heinrich, F.; Pruckner, M. Virtual experiments for battery state of health estimation based on neural networks and in-vehicle data. *J. Energy Storage* **2022**, *48*, 103856. [[CrossRef](#)]
75. Ozkurt, C.; Camci, F.; Atamuradov, V.; Odorry, C. Integration of sampling based battery state of health estimation method in electric vehicles. *Appl. Energy-Barking Oxf.* **2016**, *175*, 356–367. [[CrossRef](#)]
76. Micea, M.V.; Ungurean, L.; Cârstoiu, G.N.; Groza, V. Online state-of-health assessment for battery management systems. *IEEE Trans. Instrum. Meas.* **2011**, *60*, 1997–2006. [[CrossRef](#)]
77. Zhang, J.; Lee, J. A review on prognostics and health monitoring of Li-ion battery. *J. Power Sources* **2011**, *196*, 6007–6014. [[CrossRef](#)]
78. Ouyang, M.; Feng, X.; Han, X.; Lu, L.; Li, Z.; He, X. A dynamic capacity degradation model and its applications considering varying load for a large format Li-ion battery. *Appl. Energy* **2016**, *165*, 48–59. [[CrossRef](#)]
79. Zou, Y.; Lib, S.E.; Shao, B.; Wang, B. State-space model with non-integer order derivatives for lithium-ion battery. *Appl. Energy* **2016**, *161*, 330–336. [[CrossRef](#)]
80. Blanke, H.; Bohlen, O.; Buller, S.; De Doncker, R.W.; Fricke, B.; Hammouche, A.; Linzen, D.; Thele, M.; Sauer, D.U. Impedance measurements on lead-acid batteries for state-of-charge, state-of-health and cranking capability prognosis in electric and hybrid electric vehicles. *J. Power Sources* **2005**, *144*, 418–425. [[CrossRef](#)]
81. Barré, A.; Deguilhem, B.; Grolleau, S.; Gérard, M.; Suard, F.; Riu, D. A review on lithium-ion battery ageing mechanisms and estimations for automotive applications. *J. Power Sources* **2013**, *241*, 680–689. [[CrossRef](#)]

82. Jin, F.; Yongling, H.; Guofu, W. Comparison study of equivalent circuit model of Liion battery for electrical vehicles. *Res. J. Appl. Sci. Eng. Technol.* **2013**, *6*, 3756–3759. [\[CrossRef\]](#)
83. Rahimian, S.K.; Tang, Y. A practical data-driven battery state-of-health estimation for electric vehicles. *IEEE Trans. Ind. Electron.* **2022**, *70*, 1973–1982. [\[CrossRef\]](#)
84. Tian, N.; Wang, Y.; Chen, J.; Fang, H. On-shot parameter identification of the thevenin's model for batteries: Methods and validation. *J. Energy Storage* **2020**, *29*, 101282. [\[CrossRef\]](#)
85. Li, A.G.; Mayilvahanan, K.; West, A.C.; Preindl, M. Discrete-time modeling of Li-ion batteries with electrochemical overpotentials including diffusion. *J. Power Sources* **2021**, *500*, 229991. [\[CrossRef\]](#)
86. Arachchige, B.; Perinpanayagam, S.; Jaras, R. Enhanced prognostic model for lithium ion batteries based on particle filter state transition model modification. *Appl. Sci.* **2017**, *7*, 1172. [\[CrossRef\]](#)
87. Vanem, E.; Salucci, C.B.; Bakdi, A.; Alnes, Ø.Å. Data-driven state of health modelling-A review of state of the art and reflections on applications for maritime battery systems. *J. Energy Storage* **2021**, *43*, 103158. [\[CrossRef\]](#)
88. Plett, G.L. Sigma-point Kalman filtering for battery management systems of LiPBbased HEV battery packs Part 2: Simultaneous state and parameter estimation. *J. Power Sources* **2006**, *161*, 1369–1384. [\[CrossRef\]](#)
89. He, H.; Xiong, R.; Guo, H. Online estimation of model parameters and state-ofcharge of LiFePO₄batteries in electric vehicles. *Appl. Energy* **2012**, *89*, 413–420. [\[CrossRef\]](#)
90. Zou, Y.; Hu, X.; Ma, H.; Li, S.E. Combined State of Charge and State of Health estimation over lithium-ion battery cell cycle lifespan for electric vehicles. *J. Power Sources* **2015**, *273*, 793–803. [\[CrossRef\]](#)
91. Wang, Q.; Jiang, Y.; Lu, Y. State of health estimation for lithium-ion battery based on D-UKF. *Int. J. Hybrid. Inf. Technol.* **2015**, *8*, 16. [\[CrossRef\]](#)
92. Xiong, R.; Zhang, Y.; He, H.; Zhou, X.; Pecht, M.G. A double-scale, particle-filtering, energy state prediction algorithm for lithium-ion batteries. *IEEE Trans. Ind. Electron.* **2017**, *65*, 1526–1538. [\[CrossRef\]](#)
93. Wei, Z.; Zou, C.; Leng, F.; Soong, B.H.; Tseng, K.J. Online model identification and state-of-charge estimate for lithium-ion battery with a recursive total least squares-based observer. *IEEE Trans. Ind. Electron.* **2017**, *65*, 1336–1346. [\[CrossRef\]](#)
94. Doyle, M.; Fuller, T.F.; Newman, J. Modeling of galvanostatic charge and discharge of the lithium/polymer/insertion cell. *J. Electrochem. Soc.* **1993**, *140*, 1526–1533. [\[CrossRef\]](#)
95. Fuller, T.F.; Doyle, M.; Newman, J. Simulation and optimization of the dual lithium ion insertion cell. *J. Electrochem. Soc.* **1994**, *141*, 1–10. [\[CrossRef\]](#)
96. Zheng, L.; Zhang, L.; Zhu, J.; Wang, G.; Jiang, J. Co-estimation of state-of-charge, capacity and resistance for lithium-ion batteries based on a high-fidelity electrochemical model. *Appl. Energy* **2016**, *180*, 424–434. [\[CrossRef\]](#)
97. Bartlett, A.; Marcicki, J.; Onori, S.; Rizzoni, G.; Yang, X.G.; Miller, T. Model-based state of charge estimation and observability analysis of a composite electrode lithium-ion battery. In Proceedings of the 52nd IEEE Conference on Decision and Control, Florence, Italy, 10–13 December 2013; Volume 24, pp. 7791–7796.
98. Zou, C.; Manzie, C.; Ne, D. Multi-time-scale observer design for state-of-charge and state-of-health of a lithium-ion battery. *J. Power Sources* **2016**, *335*, 121–130. [\[CrossRef\]](#)
99. Shen, S.; Sadoughi, M.; Li, M.; Wang, Z.D.; Hu, C. Deep convolutional neural networks with ensemble learning and transfer learning for capacity estimation of lithium-ion batteries. *Appl. Energy* **2020**, *260*, 114296. [\[CrossRef\]](#)
100. Feng, X.; Weng, C.; He, X.; Han, X.; Lu, L.; Ren, D.; Ouyang, M. Online state-of-health estimation for Li-ion battery using partial charging segment based on support vector machine. *IEEE Trans. Veh. Technol.* **2019**, *68*, 8583–8592. [\[CrossRef\]](#)
101. Tan, X.; Tan, Y.; Zhan, D.; Yu, Z.; Fan, Y.; Qiu, J.; Li, J. Real-time state-of-health estimation of lithium-ion batteries based on the equivalent internal resistance. *IEEE Access* **2020**, *8*, 56811–56822. [\[CrossRef\]](#)
102. Li, X.; Yuan, C.; Wang, Z. State of health estimation for Li-ion battery via partial incremental capacity analysis based on support vector regression. *Energy* **2020**, *203*, 117852. [\[CrossRef\]](#)
103. Naha, A.; Han, S.; Agarwal, S.; Guha, A.; Khandelwal, A.; Tagade, P.; Hariharan, K.S.; Kolake, S.M.; Yoon, J. An incremental voltage difference based technique for online state of health estimation of Li-ion batteries. *Sci. Rep.* **2020**, *10*, 9526. [\[CrossRef\]](#) [\[PubMed\]](#)
104. Rezvani, M.; PhD, M.A.; Lee, S.; Lee, J.; Ni, J. *A Comparative Analysis of Techniques for Electric Vehicle Battery Prognostics and Health Management (PHM)*; SAE Technical Paper Series; SAE International: Chicago, IL, USA, 2011.
105. Andre, D.; Nuhic, A.; Soczka-Guth, T.; Sauer, D.U. Comparative study of a structured neural network and an extended Kalman filter for state of health determination of lithium-ion batteries in hybrid electric vehicles. *Eng. Appl. Artif. Intell.* **2013**, *26*, 951–961. [\[CrossRef\]](#)
106. Park, S.; Zhang, D.; Moura, S. Hybrid electrochemical modeling with recurrent neural networks for li-ion batteries. *Adv. Comput. Commun. IEEE* **2017**, *68*, 3777–3782.
107. You, G.-W.; Park, S.; Oh, D. Diagnosis of electric vehicle batteries using recurrent neural networks. *IEEE Trans. Ind. Electron.* **2017**, *64*, 4885–4893. [\[CrossRef\]](#)

108. Tan, Y.; Zhao, G. Transfer Learning With Long Short-Term Memory Network for State-of-Health Prediction of Lithium-ion Batteries. *IEEE Trans. Ind. Electron.* **2020**, *67*, 8723–8731. [\[CrossRef\]](#)
109. Li, W.; Sengupta, N.; Dechent, P.; Howey, D.; Annaswamy, A.; Sauer, D.U. Online capacity estimation of lithium-ion batteries with deep long short-term memory networks. *J. Power Sources* **2021**, *482*, 228863. [\[CrossRef\]](#)
110. Pan, H.; Lü, Z.; Wang, H.; Wei, H.; Chen, L. Novel battery state-of-health online estimation method using multiple health indicators and an extreme learning machine. *Energy* **2018**, *160*, 466–477. [\[CrossRef\]](#)
111. Shu, X.; Li, G.; Zhang, Y.; Shen, J.; Chen, Z.; Liu, Y. Online diagnosis of state of health for lithium-ion batteries based on short-term charging profiles. *J. Power Sources* **2020**, *471*, 228478. [\[CrossRef\]](#)
112. Liu, W.; Xu, Y.; Feng, X. A hierarchical and flexible data-driven method for online state-of-health estimation of Li-ion battery. *IEEE Trans. Veh. Technol.* **2020**, *69*, 14739–14748. [\[CrossRef\]](#)
113. Liu, W.; Xu, Y. Data-driven online health estimation of Li-ion batteries using a novel energy-based health indicator. *IEEE Trans. Energy Convers.* **2020**, *35*, 1715–1718. [\[CrossRef\]](#)
114. Tagade, P.; Hariharan, K.S.; Ramachandran, S.; Khandelwal, A.; Naha, A.; Kolake, S.M.; Han, S.H. Deep Gaussian process regression for lithium-ion battery health prognosis and degradation mode diagnosis. *J. Power Sources* **2020**, *445*, 227281. [\[CrossRef\]](#)
115. Yang, D.; Zhang, X.; Pan, R.; Wang, Y.; Chen, Z. A novel Gaussian process regression model for state-of-health estimation of lithium-ion battery using charging curve. *J. Power Sources* **2018**, *384*, 387–395. [\[CrossRef\]](#)
116. Li, X.; Yuan, C.; Wang, Z. Multi-time-scale framework for prognostic health condition of lithium battery using modified Gaussian process regression and nonlinear regression. *J. Power Sources* **2020**, *467*, 228358. [\[CrossRef\]](#)
117. Sheng, H.M.; Liu, X.; Bai, L.B.; Dong, H.C.; Cheng, Y.H. Small sample state of health estimation based on weighted Gaussian process regression. *J. Energy Storage* **2021**, *41*, 102816. [\[CrossRef\]](#)
118. Deng, Z.; Hu, X.; Lin, X.; Xu, L.; Che, Y.; Hu, L. General discharge voltage information enabled health evaluation for lithium-ion batteries. *IEEE/ASME Trans. Mechatron.* **2021**, *26*, 1295–1306. [\[CrossRef\]](#)
119. Meng, J.H.; Cai, L.; Stroe, D.I.; Ma, J.P.; Luo, G.Z.; Teodorescu, R. An optimized ensemble learning framework for lithium-ion Battery State of Health estimation in energy storage system. *Energy* **2020**, *206*, 118140. [\[CrossRef\]](#)
120. Gou, B.; Xu, Y.; Feng, X. State-of-health estimation and remaining-useful-life prediction for lithium-ion battery using a hybrid data-driven method. *IEEE Trans. Veh. Technol.* **2020**, *69*, 10854–10867. [\[CrossRef\]](#)
121. Li, Y.; Zhong, S.; Zhong, Q.; Shi, K. Lithium-ion battery state of health monitoring based on ensemble learning. *IEEE Access* **2019**, *7*, 8754–8762. [\[CrossRef\]](#)
122. Jorge, I.; Mesbahi, T.; Samet, A.; Boné, R. Time Series Feature extraction for Lithium-Ion batteries State-of-Health prediction. *J. Energy Storage* **2023**, *59*, 106436. [\[CrossRef\]](#)
123. Liu, K.; Shang, Y.; Ouyang, Q.; Widanage, W.D. A data-driven approach with uncertainty quantification for predicting future capacities and remaining useful life of lithium-ion battery. *IEEE Trans. Ind. Electron.* **2021**, *68*, 3170–3180. [\[CrossRef\]](#)
124. Li, W.; Sengupta, N.; Dechent, P.; Howey, D.; Annaswamy, A.; Sauer, D.U. One-shot battery degradation trajectory prediction with deep learning. *J. Power Sources* **2021**, *506*, 230024. [\[CrossRef\]](#)
125. Lei, R. A data-driven auto-CNN-LSTM prediction model for lithium-ion battery remaining useful life. *IEEE Trans. Ind. Inform.* **2021**, *17*, 3478–3487.
126. Audin, P.; Jorge, I.; Mesbahi, T.; Samet, A.; De Bertrand De Beuvron, F.; Bone, R. Auto-encoder LSTM for Li-ion SOH prediction: A comparative study on various benchmark datasets. In Proceedings of the 20th IEEE International Conference on Machine Learning and Applications, ICMLA 2021, Virtual, 13–16 December 2021; pp. 1529–1536.
127. Li, Q.; Li, D.; Zhao, K.; Wang, L.; Wang, K. State of health estimation of lithium-ion battery based on improved ant lion optimization and support vector regression. *J. Energy Storage* **2022**, *50*, 104215. [\[CrossRef\]](#)
128. Song, Y.; Liu, D.; Liao, H.; Peng, Y. A hybrid statistical data-driven method for on-line joint state estimation of lithium-ion batteries. *Appl. Energy* **2020**, *261*, 114408. [\[CrossRef\]](#)
129. Lin, M.; Yan, C.; Wang, W.; Dong, G.; Meng, J.; Wu, J. A data-driven approach for estimating state-of-health of lithium-ion batteries considering internal resistance. *Energy* **2023**, *277*, 127675. [\[CrossRef\]](#)
130. Hong, J.; Wang, Z.; Chen, W.; Wang, L.; Lin, P.; Qu, C. Online accurate state of health estimation for battery systems on real-world electric vehicles with variable driving conditions considered. *J. Clean. Prod.* **2021**, *294*, 125814. [\[CrossRef\]](#)
131. Ma, J.; Chen, B.B.; Wang, L.L.; Cui, G.L. Progress and prospect on failure mechanisms of solid-state lithium batteries. *J. Power Sources* **2018**, *392*, 94–115. [\[CrossRef\]](#)
132. Zhang, J.P.; Zhu, J.; Zhao, R.Q.; Liu, J.; Song, X.C.; Xu, N.; Liu, Y.S.; Zhang, H.T.; Wan, X.J.; Ma, Y.F.; et al. An all-in-one free-standing single-ion conducting semi-solid polymer electrolyte for high-performance practical Li metal batteries. *Energy Environ. Sci.* **2024**, *17*, 7119–7128. [\[CrossRef\]](#)
133. Zeng, Q.H.; Lu, Y.; Chen, P.P.; Li, Z.F.; Wen, X.; Wen, W.; Liu, Y.; Zhang, S.P.; Zhao, H.L.; Zhou, H.H.; et al. Semi-interpenetrating-network all-solid-state polymer electrolyte with liquid crystal constructing efficient ion transport channels for flexible solid lithium-metal batteries. *J. Energy Chem.* **2022**, *67*, 157–167. [\[CrossRef\]](#)

134. Homann, G.; Stolz, L.; Neuhaus, K.; Winter, M.; Kasnatscheew, J. Effective Optimization of High Voltage Solid-State Lithium Batteries by Using Poly(ethylene oxide)-Based Polymer Electrolyte with Semi-Interpenetrating Network. *Adv. Func. Mater.* **2020**, *30*, 2006289. [[CrossRef](#)]
135. Sakuda, A. Favorable composite electrodes for all-solid-state batteries. *J. Ceram. Soc. Jpn.* **2018**, *126*, 675–683. [[CrossRef](#)]
136. Celen, B.; Aras, A.C.; Dohr, M. Equivalent Circuit Modeling of All-Solid-State Battery by using DC-IR Data. In Proceedings of the 15th Annual IEEE International Systems Conference (SYSCON 2021), Virtual, 14 April–15 May 2021.

Disclaimer/Publisher’s Note: The statements, opinions and data contained in all publications are solely those of the individual author(s) and contributor(s) and not of MDPI and/or the editor(s). MDPI and/or the editor(s) disclaim responsibility for any injury to people or property resulting from any ideas, methods, instructions or products referred to in the content.